

コヒーレントイジングマシンによる 基底・低エネルギーイジングスピン配置の効率的サンプリング

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量子雑音の存在下における測定フィードバック型コヒーレントイジングマシン (MFB-CIM) の非線形確率ダイナミクスを利用して、イジングモデルの縮退した基底スピン配置と低エネルギースピン配置をサンプリングできることを示す。系のしきい値以上で生じる非線形ダイナミクスを忠実に捉える、一般的な離散時間ガウス状態モデルを定式化する。このモデルは、量子雑音を無視する平均場モデルと、長い光子寿命を仮定する連続時間モデルの両方の限界を克服する。数値シミュレーションによれば、光子寿命が短く (共振器フィネスが低く)、量子雑音が支配的な領域でMFB-CIMを動作させると、ホモダイン観測によって低エネルギーのイジングスピン配置を効率よく生成できる。必要な周回数は、従来の高フィネス・連続時間モデルが示唆する値よりはるかに少ない。パラメトリック駆動を停止しても、あるいはその符号を逆転しても、サンプリング性能は維持され、場合によっては向上する。一方、光学的非線形性がなければ性能は著しく低下する。辺の重みが正負の二値をとるMAX-CUT問題群では、縮退をすべて含め、イジングエネルギーの第一励起準位までの全スピン配置を十分にサンプリングする周回数は、問題サイズNに対して 1.08^N で増大する。N = 100では、各問題に対象配置が数十個 (中央値約20個) 存在し、十分なサンプリングに要する周回数の中央値は 6×10^6 であった。繰り返し周波数10 GHzの実験的MFB-CIMでは、これは実時間60 msに相当する。

I. はじめに

数十年にわたり、イジングモデルは物理学と計算科学をつなぐ重要な概念的架け橋となってきた。多くの重要な組合せ最適化問題は、イジングモデルの基底状態を求める問題へ効率的に写像できる[1]。一方、モデルの形式が単純かつ極めて一般的であるため、イジング型相互作用は多様な系に広く存在する[2]。形式的には、イジングモデルはスピン $\sigma_i = \pm 1$ の集合からなり、その配置エネルギーはイジングハミルトニアン $-\sum_{i \neq j} J_{ij} \sigma_i \sigma_j$ で与えられる。一般に、このエネルギーを最小にするスピン配置を求めるイジング問題は、現在の従来型コンピュータでは計算が困難である[3]。このため、物理的なイジング型システムを、組合せ最適化などに取り組む専用計算ハードウェアとして利用することに大きな関心が集まっている。マイクロ波超伝導回路による量子アニーラ[4, 5]から、非線形光発振器のネットワークに基づくコヒーレントイジングマシン[6–9]まで、さまざまな方式の研究が進められている[10–15]。

組合せ最適化では、イジングスピンの基底配置を一つ見つけることに重点が置かれる場合が多い。しかし多くの応用では、縮退した基底配置の多数またはすべてを得ることが望ましく、

場合によっては、多数の低エネルギー配置もサンプリングしたい[16]。このような能力は、イジングモデルのスピン配置に関する分布情報を得る応用で特に有用である。例えば、イジング型相互作用をもつ物理シミュレーションの基底状態エントロピーの推定や、機械学習の生成モデルとしてのボルツマンマシンの実装が挙げられる[17–19]。産業用途では、最適化問題の候補解を複数利用できれば、プロセスをより効率的で柔軟にできる。例えば創薬[20–23]の構造に基づくリード最適化では、同時に試験する複数の候補分子を生成できる。近年、大規模な最適化問題を別々に解く部分問題へ分割する場合 (例えばハードウェアの制約に対応する場合)、各部分問題の最適解だけでなく複数の低エネルギー標本を使うことで、元の問題に対するより良い解を構成できることも指摘された[24, 25]。しかし、組合せ最適化向けのイジングソルバが、すべての基底状態や低エネルギー状態のサンプリングにも適しているとは限らない。例えばD-Wave Systemsの商用量子アニーラはイジング問題の基底状態探索で成果を上げているが、その動作原理によって、縮退基底状態の標本分布に指数関数的な偏りが生じ得る[26–28]。こうした問題はソルバを構成するハードウェアの詳細と本質的に結び付くことが多い。そのため、サンプリング性能を数値的に調べるには、装置とその動作を正確に表すモデルを開発し、解析する必要がある。

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本論文では、測定フィードバック型コヒーレントイジングマシン (MFB-CIM) [7-9, 29] のサンプリング性能を調べる。これは、縮退光パラメトリック発振器 (DOPO) のネットワークを用いてイジング最適化を行うために考案されたハードウェアである。リアルタイムの測定フィードバックによって、イジング相互作用をネットワークのダイナミクスに組み込む。特にガウス状態モデルを用い、MFB-CIMのダイナミクスの中で量子雑音がどのように生じるかを調べ、確率的な非線形ダイナミクスが低エネルギーのイジング配置の効率的なサンプリングを促進できるかを検討する。MFB-CIMの完全量子的な取り扱いも可能である[29]が、組合せ最適化やサンプリングに関係する大規模系の数値研究は、量子相関を二次まで(観測の共分散まで)考慮するガウス状態領域に限られる[30, 31]。この近似は、これまでに知られているすべての実験的MFB-CIMの動作領域と整合する[7-9]。同時に、スクイーピング・反スクイーピング、測定の不確実性と反作用といった、量子雑音による重要な現象を正確に扱うことができる[29, 32, 33]。これらは本研究のサンプリング性能に中心的な役割を果たすが、平均場モデルでは通常無視される。

MFB-CIMが縮退した基底スピン配置と低エネルギー励起スピン配置の標本を生成できる可能性は、連続時間で定式化されたガウス状態モデル[35]を用いた文献[34]で最近指摘された。本研究で示すように、このような連続時間モデルは、構成要素であるDOPOの共振器減衰時間が他のすべての系の時間スケールより長い高フィネス極限では、MFB-CIMのダイナミクスを正しく捉える。一方、低フィネス系は本来帯域幅が広いと、少なくとも原理的には計算時間を大幅に短縮できる。実際、光結合型と測定フィードバック型のどちらでも、CIMの実験実装の多くは、共振器減衰時間が短い低フィネス領域で動作するDOPOを用いている[6-9]。低フィネス系は、系の状態に対する一連の離散的な操作としてダイナミクスを表す離散時間の記述が適している[36]。MFB-CIMの離散時間での量子的扱いは、理論的には文献[37, 38]で研究されている。後者は非ガウス量子状態モデルを用い、小さな問題サイズでしか数値的に扱えなかった。前者はガウス状態モデルによってMFB-CIMの線形ダイナミクスを研究したが、しきい値付近やしきい値以上で重要な役割を果たし得る非線形利得飽和は、現象論的にしか考慮していなかった。これらの制約を回避するため、本研究では、非線形利得飽和を物理的にモデル化した離散時間ガウス状態量子モデルを開発する。これにより、低・中フィネスのMFB-CIMを、しきい値より下からしきい値を経てその上まで調べることができる。著者らの知る限り、

ここで示すモデルは、ガウス状態領域で動作する大規模MFB-CIMを数値シミュレーションするための、現時点で最も一般的な取り扱いである。

II. MFB-CIMの離散時間 ガウス量子モデル

概念的には、コヒーレントイジングマシン (CIM) はN個の縮退光パラメトリック発振器 (DOPO) からなる系である。DOPOは、飽和する位相感応利得を示す非線形光発振器である。発振しきい値未満で励起すると、その状態は直交位相成分がスクイーズされた真空状態によく記述される。しきい値を十分に超えると、ポンプ枯渇による利得の非線形飽和が、二つの位相双安定な明るいコヒーレント状態(0位相状態と π 位相状態)のいずれかに系を安定化する。DOPOネットワークにイジングスピンを符号化するため、これらの双安定位相状態を、それぞれ $\sigma = \pm 1$ に対応させる。DOPO間の相互作用を設計すれば、所望のイジング結合行列Jに支配されるダイナミクスを実現できる。

図1は、特に成功を収めているCIMの実験実装の概略である。単一の光ファイバループ共振器内で、同期励起された時間多重「信号」パルスとしてDOPOを実現し、同期したリアルタイム測定フィードバックによってパルス間相互作用を媒介する。この測定フィードバック型CIM (MFB-CIM) では、信号パルスの時間間隔は $1/f_{\text{rep}}$ であり、N個のパルスを収容する共振器長は約 Nc/f_{rep} となる。共振器を1周するたびに、信号パルスは、光源レーザーの第二高調波から得られ外部注入された同期ポンプパルスとともに、非線形 $\chi^{(2)}$ 結晶を順次伝搬する。これにより同相q成分に沿った位相感応増幅が生じる。次に、出力結合器で信号パルスの一部を順次取り出す。出力をq成分のホモダイン検出器で測定すると、共振器内信号パルスのq成分振幅を間接的かつ弱く測定できる。ここで重要なのは、N個のホモダイン測定値の符号配置、例えば $(\text{sgn } w_1, \dots, \text{sgn } w_N)$ が、 $\sigma_j \leftrightarrow \text{sgn } w_j$ という対応の下で、サンプリングされたイジングスピン配置を構成することである。最後に、パルス間相互作用を実装するため、FPGAがホモダイン測定結果を受け取り、フィードバック信号 $v_i \propto \sum_{j=1}^N J_{ij} w_j$ を計算する。この信号は、光源レーザーから生成され、その強度と位相を v_i で定めたフィードバックパルス(例えば同期光変調器を用いる)を外から同期注入することで、対応する第i信号パルスに作用する。注入パルスと共振器内信号パルスの干渉が、より低いエネルギーのイジングスピン配置へと系を導き、MFB-CIM内で結合行列Jを動的に実現する。

このようにイジング結合の構造をダイナミクスへ埋め込むと、状態の時間発展は、次の三つの一般的な要素の相互作用に支配される。

(i) 信号振幅を双安定なスピン値へ導く非線形性、(ii) イジングエネルギーを最小化する集団配置へ系を導く線形結合、(iii) 弱いホモダイン測定に内在する不確定性と、それに続く測定反作用・フィードバック注入から生じ、

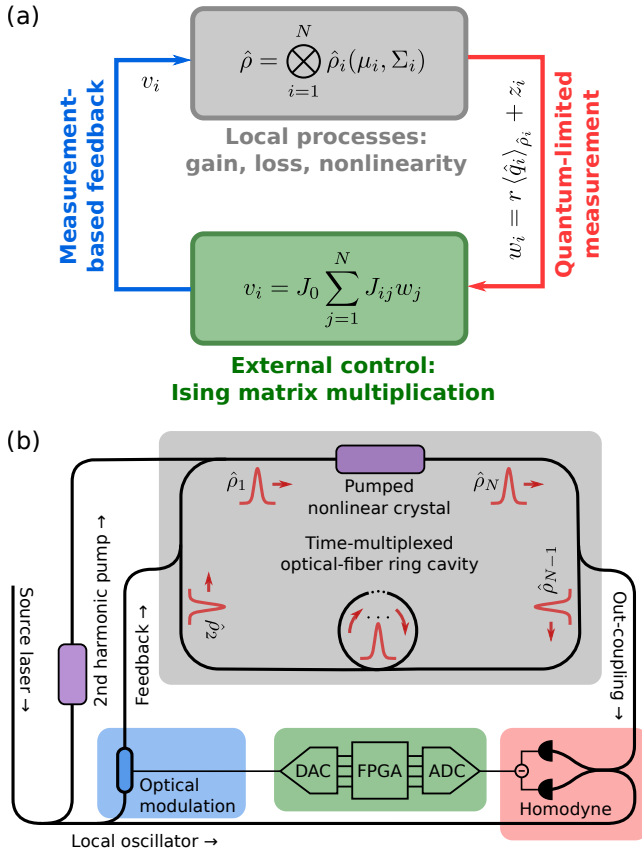


図1. 測定フィードバック型コヒーレントイジングマシン (MFB-CIM) の概略。(a) 抽象的には、MFB-CIMはN個の光学モードからなる系（灰色の枠）であり、各モードの状態 ρ_i は平均ベクトル μ_i と共分散行列 Σ_i で特徴付けられる。各モードは光利得・損失・非線形性の作用を受ける。系を弱測定で調べることで、正規分布（ガウス分布）に従う量子雑音 z_i を伴う、同相成分 q_i の推定値 w_i を得る。この推定値を外部制御器（緑色の枠）が処理し、指定されたイジング問題の行列 J_{ij} に基づくフィードバック信号 v_i を生成する。こうしてループを閉じると、イジング結合が系のダイナミクスに組み込まれる。(b) 時間多重方式[7, 8]によるMFB-CIMの光学構成。光パルス ρ_i が共振器内を循環し、励起された非線形結晶が利得と非線形性を与え、さらに付加的な線形損失が存在する。出力結合器で各パルスの一部を取り出して平衡光ホモダイン測定を行い、電気信号 w_i を得る。FPGAはこの信号を処理してフィードバック v_i を生成する。局部発振光の強度・位相を変調して光信号に再符号化し、共振器へ再注入することでパルスを光学的に変位させ、ループを閉じる。

系のダイナミクスに確率性をもたらす量子雑音、である。図2に概念的に示すように、信号振幅は確率的かつ非線形に発展するが、イジング結合行列で決まる低エネルギーの符号配置を強く選好する。連続時間極限では、この確率軌道を、量子雑音に駆動されたポテンシャル地形上の勾配降下と捉えると便利で直感的である。状態が時間とともに確率的に発展すると、各スピンが見る瞬間的なポテンシャルも動的に変わり、非線形性と結合の相互作用によって生じる局所極小の周囲や極小間へ系を導く。本論文ではこのエネルギー地形の概念そのものの解明には重点を置かないが、量子雑音が支配的な領域のMFB-CIMを多安定系として捉える描像は、以下の定式化に有用な物理的直感を与える。

MFB-CIMのように、単一光子が引き起こす非線形性が比較的弱く、連続ホモダイン測定を受ける開放散逸ボース系は、しきい値における古典的分岐を通過して発展する場合でも、通常ガウス状態をよく近似できる。形式的には、ガウス状態近似とは、測定結果で条件付けた量子状態のウィグナー関数がガウス分布でよく近似されることを意味する。したがって、平均場振幅の集合と、その量子相関および不確定性を記述する共分散の集合だけで、状態を完全に特徴付けられる[30, 31]。もう一つの非常に有用な簡略化は、特にMFB-CIMに適用されるもので、測定フィードバックを介して相互作用していても、パルス間に量子もつれがないことである。図1の物理過程は信号パルス間の局所操作と古典通信 (LOCC) だけで構成されるため、異なる信号パルス間の共分散はゼロとなる。

図1に含まれる出力結合、ホモダイン測定、フィードバック注入などの多くは線形操作である。ガウス状態については、各光学素子を通過するときに信号パルスが離散的な変換を受けるとする離散時間形式を用いると、これらの操作を特に簡潔に記述できる。この方法は、振幅が有効ハミルトニアンと、連続的な損失・測定を表すリウヴィル超演算子の下で連続的に発展する、量子光学の従来の連続時間モデルとは異なる。もちろん図1の構成は時間多重されたパルス系であるため、パルス幅がパルス間隔に比べて短い場合には、自然に離散時間モデルに適している。共振器フィネスが高いとき、連続時間モデルは離散時間モデルの適切な近似と考えられる。この極限では、1周あたりの利得・損失・測定の効果がすべて小さく、共振器内の状態は各周回でわずかしき変化しない。そのため系全体のダイナミクスは、

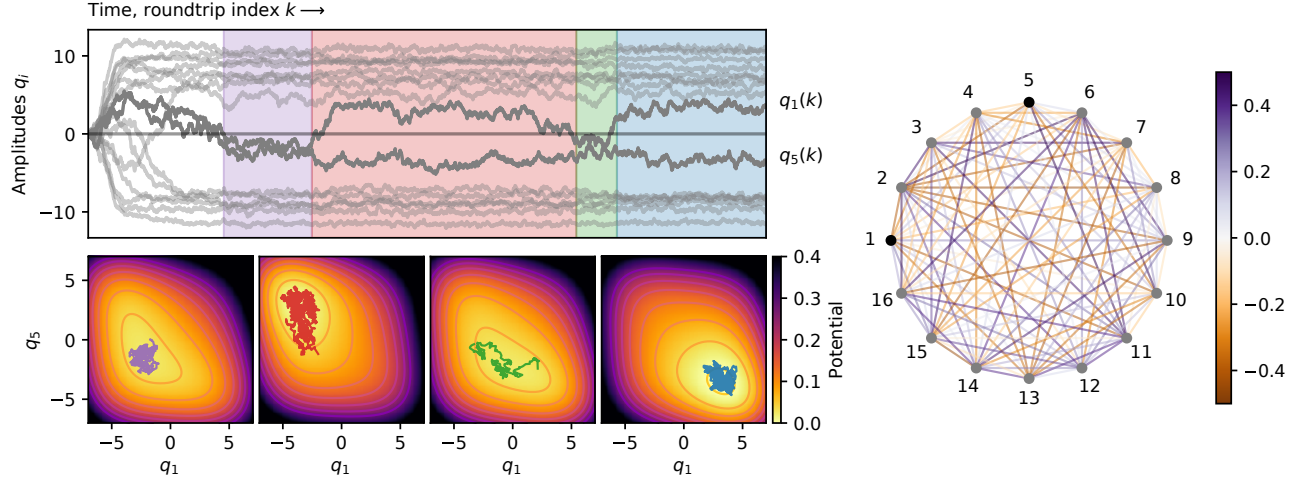


図2. 量子雑音がMFB-CIMの非線形ダイナミクスとスピン配置のサンプリングに及ぼす影響の概念図。右側のグラフは、代表的なイジング問題の一例を示す。左上は、本論文II節の離散時間ガウスモデルに従う、第*i*パルスの q 成分の期待値 q_i の確率的時間発展である。このモデルの連続時間極限では、これらのダイナミクスは N 次元ポテンシャル上の勾配降下として直感的に理解できる。例えば式(24a)から、 $V(q) = -\sum_{i=1}^N [(p - \kappa - \gamma)q_i^2/2 - gq_i^4/8 + (\lambda/2)\sum_{j=1}^N J_{ij}q_iq_j]$ となる。上段で色分けした各時間窓について、左下に系の軌道を (q_1, q_5) 座標へ射影した二次元断面を示す。対応する $V(q)$ の断面は、他のすべての座標で決まる瞬時値を、その時間窓内で平均して可視化した。一連の図は、量子雑音が状態にランダムな揺らぎを生じさせ、 $V(q)$ が異なる符号配置へ確率的に系を導くことで、この問題のさまざまな低エネルギーイジング配置をサンプリングする様子を表している。

粗視化した連続時間の微分方程式でよく記述される。しかし、共振器フィネス、したがって装置のダイナミクスの帯域幅を変えたときのサンプリング性能への影響を調べるには、より一般的な離散時間形式が必要になる。

MFB-CIMの操作のうち、離散時間ガウスモデルで容易に扱えないのは、信号パルスの結晶内伝搬である。しきい値未満では、この操作は q 成分に沿った線形の位相感応利得でよく記述され[32]、文献[37]では離散時間のスキージング操作としてモデル化されている。一方、DOPOがしきい値付近またはそれ以上にあると、共振搬するポンパルスが枯渇し、利得が飽和して非線形ダイナミクスが生じる。単一光子による量子非線形性が強ければ、ガウス状態近似を超える物理現象さえ現れる[38]。以下のモデルの主な貢献は、ガウス状態領域における利得飽和を数値的に効率よく扱う方法を与えることである。これにより、離散時間ガウスモデルをしきい値以上へ拡張できる。同時に、高フィネス極限では、MFB-CIMの標準的な量子光学モデルから導かれる連続時間ガウス状態モデル[35]との整合性も保たれる。

II A節では、MFB-CIMのすべての線形操作を簡潔に記述できるガウス状態の基本形式を概説する。II B節では、結晶内の非線形伝搬についてガウス状態の運動方程式を導く。

続いてII C節では、MFB-CIMの状態を1周回分発展させる反復手順全体をまとめ、離散時間の動力学モデルを完成させる。最後にII D節で、離散時間モデルが従来のMFB-CIMの連続時間モデルにどのように帰着し、接続されるかを概説する。

A. 基本形式

時間多重MFB-CIMを、消滅演算子 $\hat{a}_i$ が $[\hat{a}_i, \hat{a}_i^\dagger] = \delta_{ij}$ を満たす N モードのボース系として抽象化する。直交位相演算子 $z := (q_1, p_1, \dots, q_N, p_N)$ は、 $[z_k, z_\ell] = i\Omega_{k\ell}$ を満たすように定義する。ここで $\Omega := \oplus_{i=1}^N [[0, 1], [-1, 0]]$ はシンプレクティック形式である。ガウス状態[30, 31, 39, 40]は、平均ベクトル μ と共分散行列 Σ のみで完全に定まる。量子状態を $\rho(\mu, \Sigma)$ と書くと、その一次モーメント（平均ベクトル）は

$$\mu_k := \text{tr}(\hat{z}_k \hat{\rho}), \quad (1a)$$

二次モーメント（共分散行列）は

$$\Sigma_{k\ell} := \text{tr} \left(\frac{1}{2} (\delta \hat{z}_k \delta \hat{z}_\ell + \delta \hat{z}_\ell \delta \hat{z}_k) \hat{\rho} \right), \quad (1b)$$

ここで $\delta z := z - \mu$ は各直交位相成分の揺らぎ演算子のベクトルである。さらにMFB-CIMでは、

LOCCダイナミクスにより量子もつれが生じないため、次の簡略化ができる。

$$\mu = \bigoplus_{i=1}^N \mu^{(i)} \quad \text{and} \quad \Sigma = \bigoplus_{i=1}^N \Sigma^{(i)}, \quad (2a)$$

具体的には、

$$\mu^{(i)} := (\langle \hat{q}_i \rangle, \langle \hat{p}_i \rangle), \quad (2b)$$

$$\Sigma^{(i)} := \begin{pmatrix} \langle \delta \hat{q}_i^2 \rangle & \frac{1}{2} \langle \delta \hat{q}_i \delta \hat{p}_i + \delta \hat{p}_i \delta \hat{q}_i \rangle \\ \frac{1}{2} \langle \delta \hat{q}_i \delta \hat{p}_i + \delta \hat{p}_i \delta \hat{q}_i \rangle & \langle \delta \hat{p}_i^2 \rangle \end{pmatrix}, \quad (2c)$$

したがってMFB-CIMの共分散行列は、一般の場合の $O(N^2)$ 個ではなく、非ゼロ要素を高々 $4N$ 個（独立なものは $3N$ 個）しかもたない。それに対応して、予想どおり量子状態は

$$\hat{\rho}(\mu, \Sigma) = \bigotimes_{i=1}^N \hat{\rho}(\mu^{(i)}, \Sigma^{(i)}), \quad (2d)$$

と因数分解される。ここで二つのベクトル $\mu^{(1)}$ と $\mu^{(2)}$ について、 $\mu^{(1)} \oplus \mu^{(2)}$ は両者の連結を表す。二つの行列 $\Sigma^{(1)}$ と $\Sigma^{(2)}$ については、 $\Sigma^{(1)} \oplus \Sigma^{(2)}$ は対角ブロックが $\Sigma^{(1)}$ と $\Sigma^{(2)}$ 、非対角ブロックが0のブロック対角行列を表す。より一般に、状態 $\rho(\mu^{(a)}, \Sigma^{(a)})$ と $\rho(\mu^{(b)}, \Sigma^{(b)})$ の二つの系を合わせると、合成系の状態は次式で記述される。

$$\begin{aligned} \hat{\rho}(\mu^{(a)}, \Sigma^{(a)}) \otimes \hat{\rho}(\mu^{(b)}, \Sigma^{(b)}) \\ = \hat{\rho}(\mu^{(a)} \oplus \mu^{(b)}, \Sigma^{(a)} \oplus \Sigma^{(b)}). \end{aligned} \quad (3)$$

一方、 $\rho(\mu^{(a,b)}, \Sigma^{(a,b)})$ が二つのモードの合成系ならば、モードbに対応する部分空間を取り除く射影によって、bの部分トレースを取れる。

$$\text{tr}_b [\hat{\rho}(\mu^{(a,b)}, \Sigma^{(a,b)})] := \hat{\rho}(P\mu^{(a,b)}, P\Sigma^{(a,b)}P^T), \quad (4a)$$

この場合の射影行列は、

$$P := \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (4b)$$

以上で基本形式を定めた。非線形結晶内の伝搬へ進む前に、MFB-CIMの動作に必要な線形操作を簡単に説明する。基本操作は、損失と出力結合をモデル化するビームスプリッタ、フィードバックをモデル化するコヒーレント注入、そしてホモダイン測定からなる。

2モード状態 $\rho(\mu, \Sigma)$ に作用し、場の交換振幅が r （パワー交換率が r^2 ）である2モードビームスプリッタは、次のように記述できる。

$$\mathcal{B}_r[\hat{\rho}(\mu, \Sigma)] := \hat{\rho}(S\mu, S\Sigma S^T) \quad (5a)$$

ここで、ビームスプリッタ行列は

$$S := \begin{pmatrix} t & 0 & -r & 0 \\ 0 & t & 0 & -r \\ r & 0 & t & 0 \\ 0 & r & 0 & t \end{pmatrix}, \quad (5b)$$

ここで $t := \sqrt{1 - r^2}$ は自己散乱振幅である。

変位の二つの直交位相成分を表す $\alpha \in \mathbb{R}^2$ を、モードaへコヒーレントに注入する操作は、平均を α/ε だけ変位させた新しいモードbを導入し、場の交換振幅 $\varepsilon \rightarrow 0$ のビームスプリッタをaとbに作用させることで得られる。この極限ではaからbへの注入はなくなるが、bの平均が α/ε に比例するため、aが受ける変位全体は一定値へ近づく。

$$\begin{aligned} \mathcal{V}_\alpha[\hat{\rho}(\mu^{(a)}, \Sigma^{(a)})] \\ := \lim_{\varepsilon \rightarrow 0} \text{tr}_b \left(\mathcal{B}_\varepsilon[\hat{\rho}(\mu^{(a)}, \Sigma^{(a)}) \otimes \hat{\rho}(\varepsilon^{-1}\alpha^{(b)}, \Sigma_0^{(b)})] \right), \end{aligned}$$

$\Sigma_0 := \text{diag}(1/2, 1/2)$ はコヒーレント状態の共分散である。この極限の結果は単純で、簡単のため上付き添字を省くと、予想どおり次式となる。

$$\mathcal{V}_\alpha[\hat{\rho}(\mu, \Sigma)] = \hat{\rho}(\mu + \alpha, \Sigma). \quad (6)$$

最後に、2モード系 $\rho(\mu^{(a,b)}, \Sigma^{(a,b)})$ のモードbに対するq成分の測定を考える。この系は一般に次の形で書ける。

$$\mu^{(a,b)} := \begin{pmatrix} \mu^{(a)} \\ \mu^{(b)} \end{pmatrix} \quad \text{and} \quad \Sigma^{(a,b)} := \begin{pmatrix} \Sigma^{(a)} & V \\ V^T & \Sigma^{(b)} \end{pmatrix},$$

Vは二つのモード間の量子相関を表す。測定から得られるランダムな出力は、次の正規分布に従う。

$$w \sim \mathcal{N}(\mu_q^{(b)}, \Sigma_{qq}^{(b)}), \quad (7)$$

$\mu_q^{(b)}$ と $\Sigma_{qq}^{(b)}$ は、モードbのq成分の平均と分散である（qは単に第1成分を表す）。測定後、モードbは q_b の固有状態 $|q = w\rangle_b$ へ射影され、形式的には部分トレースで消去できる。モードaへの適切な反作用は、次式で表される[39, 40]。

$$\begin{aligned} \mu_w^{(a)} &:= \mu^{(a)} + V(Q\Sigma^{(b)}Q)^+((w, 0)^T - \mu^{(b)}), \\ \Sigma_w^{(a)} &:= \Sigma^{(a)} - V(Q\Sigma^{(b)}Q)^+V^T, \end{aligned}$$

$Q := \text{diag}(1, 0)$ はモードbのq成分への射影演算子であり、任意の行列Mに対し M^+ はムーア・ペンローズ擬似逆行列を表す。すなわち測定結果wを得た後のモードaの条件付き状態は、 $\rho(\mu_w^{(a)}, \Sigma_w^{(a)})$ となる。この単純な2モードの場合には、 $V = (v_q v_p)$ と書けば、別のより明示的な式が得られる。擬似逆行列を解析的に計算すると[39, 40]、

$$\mu_w^{(a)} = \mu^{(a)} + \left(\frac{w - \mu_q^{(b)}}{\Sigma_{qq}^{(b)}} \right) v_q, \quad (8a)$$

$$\Sigma_w^{(a)} = \Sigma^{(a)} - \frac{v_q v_q^T}{\Sigma_{qq}^{(b)}}. \quad (8b)$$

ホモダイン測定とその反作用を合わせた過程を、次の操作で表す。

$$\mathcal{M}_b \left[\hat{\rho}(\mu^{(a,b)}, \Sigma^{(a,b)}) \right] := \hat{\rho}(\mu_w^{(a)}, \Sigma_w^{(a)}), \quad (9)$$

これは式(7)で与えられる測定出力 w で条件付けられる。

出力結合・測定・フィードバック注入を表すこれらの線形写像は比較的単純だが、散逸的な線形損失も考慮する必要がある。実際のCIMにおける損失源は実装によって異なる。主なものには、モード整合の不完全さやフレネル反射による結晶端面損失と、鏡での散乱やファイバへの入出力時のモード整合不良による共振器内伝搬損失がある。現実的な実験では通常、結晶端面損失が支配的なので、簡単のため、すべての損失機構をまとめ、結晶の前後に置いた一対の部分透過ビームスプリッタで表す。測定用の出力結合器と同様、これらも共振器内光の一部を取り出す。ただし、出力パルスホモダイン測定して状態に反作用を与えるのではなく、この外部パルスは測定できないと仮定し、単に部分トレースで消去する。その結果、状態には散逸が生じる。

B. 非線形結晶内の伝搬

離散時間モデルで最も難しいのは、非線形結晶内のパルス伝搬である。これは動的な非ガウス過程であり、測定やフィードバックを含め、理想的にはガウス操作として扱える他の操作とは異なる。入射する第 i 信号パルスのモード $\hat{a}_i$ ごとに、コヒーレント状態に初期化した新しいポンプパルスのモード $\hat{b}$ を、ダイクロイック結合器で光路へ注入すると仮定する。ポンプは信号と同期して共伝搬し、信号とポンプの間に、次のハミルトニアンで記述されるパラメトリック相互作用を生じさせる。

$$\hat{H}_{\text{nl}}^{(i)} / \hbar = \frac{i\epsilon}{2} \left(\hat{b} \hat{a}_i^{\dagger 2} - \hat{b}^{\dagger} \hat{a}_i^2 \right), \quad (10)$$

結合率 ϵ は、結晶長と初期ポンプ振幅が与えられたときの、信号パルスが受ける小信号パラメトリック利得全体を決める。この2モード相互作用型のハミルトニアンでは、分散によるウォークオフや他のパルス変形を避けられるほどパルスが時間的に長いか、あるいは分散が十分に制御されていると仮定する。これにより、信号とポンプの両パルスを場の単一モード励起として抽象化できる。このモデルでは、時間・周波数・空間などのモード整合の不完全さを、すべて結合率 ϵ に含める。

一般にハミルトニアン(10)は、ポンプと信号の合成状態に量子もつれと非ガウス性の両方を生じさせるため、正確な記述には2モードの完全な合成ヒルベルト空間が必要になる。結晶内伝搬をガウス形式と両立させるため、高次モードで特徴付けられる状態の非ガウス性が無視できると仮定し、式(10)が生成するポンプと信号のガウスモードの運動方程式(EOM)を導く。この近似は、DOPOの飽和光子数が大きく、単一光子による利得飽和が小さい場合に有効である。運動方程式を結晶の入射端面から出射端面まで数値積分すれば、次のように抽象的に表される非線形写像が得られる。

$$\chi : \hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \otimes \hat{\rho}(\mu_0^{(b)}, \Sigma_0^{(b)}) \mapsto \hat{\rho}(\mu^{(i,b)}, \Sigma^{(i,b)}), \quad (11)$$

この写像は、コヒーレント状態のポンプと量子もつれのないうるガウス信号パルスからなる入射状態に作用し、相関したポンプ・信号のガウス状態を生成する。結晶内伝搬が終わった後には、一般に信号とともつれ得るポンプパルスをどう扱うかも決める必要がある。ここではポンプモードの部分トレースを取り、信号パルスのみを表す混合ガウス状態を得る。この信号状態の非純粋性は、二光子吸収による散逸、または同じことだが、信号からポンプへの逆変換によるエネルギー損失と解釈できる。

量子ダイナミクスをガウス部分空間に制限する直接的な方法は、式(10)から直交位相演算子のハイゼンベルク運動方程式を求め、二次までのモーメント展開を行うことである[41]。第 i 信号パルス $\hat{a}_i$ と対応するポンプパルス $\hat{b}$ の結晶内伝搬は、次式に従う。

$$\frac{d\hat{a}_i}{d\tau} = \epsilon \hat{b} \hat{a}_i^{\dagger}, \quad \frac{d\hat{b}}{d\tau} = -\frac{\epsilon}{2} \hat{a}_i^2. \quad (12)$$

便宜上、 $\hat{a}_i = \hat{x}_i + i\hat{y}_i$ 、 $\hat{b} = \hat{x}_b + i\hat{y}_b$ と書く。ここで $\hat{x}_i := \hat{q}_i / \sqrt{2}$ 、 $\hat{y}_i := \hat{p}_i / \sqrt{2}$ である。これらの規格化した直交位相演算子は、次式に従って発展する。

$$\begin{aligned} \frac{d\hat{x}_i}{d\tau} &= \epsilon (\hat{x}_b \hat{x}_i + \hat{y}_b \hat{y}_i), & \frac{d\hat{x}_b}{d\tau} &= -\frac{\epsilon}{2} (\hat{x}_i^2 - \hat{y}_i^2), \\ \frac{d\hat{y}_i}{d\tau} &= \epsilon (\hat{y}_b \hat{x}_i - \hat{x}_b \hat{y}_i), & \frac{d\hat{y}_b}{d\tau} &= -\frac{\epsilon}{2} (\hat{x}_i \hat{y}_i + \hat{y}_i \hat{x}_i). \end{aligned}$$

一次モーメントの発展は、上の式の期待値を取れば得られる。積を分解するには、 $\langle \hat{z}_1 \hat{z}_2 \rangle = \langle \delta \hat{z}_1 \delta \hat{z}_2 \rangle + \langle \hat{z}_1 \rangle \langle \hat{z}_2 \rangle$ という関係を使い、任意の二演算子の積の期待値を、平均と共分散で表せばよい。ただし、そのためには共分散の時間発展も追跡する必要がある。共分散の運動方程式を導くため、次の一般式を用いる。

$$\frac{d}{d\tau} \langle \delta \hat{z}_1 \delta \hat{z}_2 \rangle = \left\langle \frac{d\hat{z}_1}{d\tau} \hat{z}_2 + \hat{z}_1 \frac{d\hat{z}_2}{d\tau} \right\rangle - \frac{d}{d\tau} (\langle \hat{z}_1 \rangle \langle \hat{z}_2 \rangle).$$

この式を適用する際に重要なのは、次のガウスモーメントの仮定を置くことである。

$$\langle \hat{z}_1 \hat{z}_2 \hat{z}_3 \rangle \mapsto \langle \hat{z}_1 \rangle \langle \delta \hat{z}_2 \delta \hat{z}_3 \rangle + \langle \hat{z}_2 \rangle \langle \delta \hat{z}_1 \delta \hat{z}_3 \rangle + \langle \hat{z}_3 \rangle \langle \delta \hat{z}_1 \delta \hat{z}_2 \rangle + \langle \hat{z}_1 \rangle \langle \hat{z}_2 \rangle \langle \hat{z}_3 \rangle, \quad (13)$$

ここでは、三次の（非ガウス）中心モーメントについて、 $\langle \delta \hat{z}_1 \delta \hat{z}_2 \delta \hat{z}_3 \rangle = 0$ と仮定している。

この手順で導いた完全な運動方程式は付録Cに示す。一般に、 $[\delta \hat{x}, \delta \hat{y}] = i/2$ を使えば $\langle \delta \hat{x} \delta \hat{y} \rangle$ から $\langle \delta \hat{y} \delta \hat{x} \rangle$ を求められるため、追跡すべき共分散は10個である。しかし、位相感応増幅の性質を利用すると、さらに簡略化できる。初期状態が (i) $\langle \hat{y}_i \rangle = \langle \hat{y}_b \rangle = 0$ （直交相成分の変位がない）、

(ii) $\langle \{ \delta \hat{x}_i, \delta \hat{y}_i \} \rangle = \langle \{ \delta \hat{x}_b, \delta \hat{y}_b \} \rangle = \langle \delta \hat{x}_b \delta \hat{y}_i \rangle = \langle \delta \hat{y}_b \delta \hat{x}_i \rangle = 0$ （同相成分と直交相成分の揺らぎがすべて無相関）を満たすとする。線形損失と出力結合は、二つの直交位相成分に独立に作用する受動操作であり、測定とフィードバック注入はq成分にしか作用しない。したがって、初めから存在しない直交相変位や成分間相関を、線形操作が生成することはない。結晶内伝搬についても、付録Cの完全な運動方程式から、これらの条件が入射時に成り立てば伝搬中も維持されることが分かる。よって、

$$\langle \hat{y}_i \rangle = \langle \hat{y}_b \rangle = 0, \quad (14a)$$

$$\langle \frac{1}{2} \{ \delta \hat{x}_i, \delta \hat{y}_i \} \rangle = \langle \frac{1}{2} \{ \delta \hat{x}_b, \delta \hat{y}_b \} \rangle = 0, \quad (14b)$$

$$\langle \delta \hat{x}_b \delta \hat{y}_i \rangle = \langle \delta \hat{y}_b \delta \hat{x}_i \rangle = 0, \quad (14c)$$

を結晶内伝搬における不変条件とできる。

以上から最終的なガウス状態の運動方程式を得る。これを数値積分すれば、式(11)の結晶伝搬写像 χ を実装できる。平均の方程式は次のとおりである。

$$\frac{d \langle \hat{x}_i \rangle}{d\tau} = \epsilon \langle \hat{x}_b \rangle \langle \hat{x}_i \rangle + \epsilon \langle \delta \hat{x}_b \delta \hat{x}_i + \delta \hat{y}_b \delta \hat{y}_i \rangle, \quad (15a)$$

$$\frac{d \langle \hat{x}_b \rangle}{d\tau} = -\frac{\epsilon}{2} \langle \hat{x}_i \rangle^2 - \frac{\epsilon}{2} \langle \delta \hat{x}_i^2 - \delta \hat{y}_i^2 \rangle, \quad (15b)$$

一方、共分散の運動方程式は、

$$\frac{d \langle \delta \hat{x}_i^2 \rangle}{d\tau} = +2\epsilon \langle \hat{x}_b \rangle \langle \delta \hat{x}_i^2 \rangle + 2\epsilon \langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle, \quad (16a)$$

$$\frac{d \langle \delta \hat{y}_i^2 \rangle}{d\tau} = -2\epsilon \langle \hat{x}_b \rangle \langle \delta \hat{y}_i^2 \rangle + 2\epsilon \langle \hat{x}_i \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle, \quad (16b)$$

$$\frac{d \langle \delta \hat{x}_b^2 \rangle}{d\tau} = -2\epsilon \langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle, \quad (16c)$$

$$\frac{d \langle \delta \hat{y}_b^2 \rangle}{d\tau} = -2\epsilon \langle \hat{x}_i \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle, \quad (16d)$$

$$\frac{d \langle \delta \hat{x}_b \delta \hat{x}_i \rangle}{d\tau} = \epsilon \langle \hat{x}_i \rangle \langle \delta \hat{x}_b^2 - \delta \hat{x}_i^2 \rangle + \epsilon \langle \hat{x}_b \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle, \quad (16e)$$

$$\frac{d \langle \delta \hat{y}_b \delta \hat{y}_i \rangle}{d\tau} = \epsilon \langle \hat{x}_i \rangle \langle \delta \hat{y}_b^2 - \delta \hat{y}_i^2 \rangle - \epsilon \langle \hat{x}_b \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle. \quad (16f)$$

これらの初期条件も明示できる。 $\langle \hat{x}_i \rangle$ 、 $\langle \delta \hat{x}_i^2 \rangle$ 、 $\langle \delta \hat{y}_i^2 \rangle$ は当然、結晶に入射する信号パルスの状態に依存するが、次のように書ける。

$$\langle \hat{x}_i \rangle(0) = \frac{1}{\sqrt{2}} \langle \hat{q}_i \rangle, \quad \langle \hat{x}_b \rangle(0) = \frac{1}{\sqrt{2}} \beta, \quad (17a)$$

$$\langle \delta \hat{x}_i^2 \rangle(0) = \frac{1}{2} \langle \delta \hat{q}_i^2 \rangle, \quad \langle \delta \hat{y}_i^2 \rangle(0) = \frac{1}{2} \langle \delta \hat{p}_i^2 \rangle, \quad (17b)$$

$$\langle \delta \hat{x}_b^2 \rangle(0) = \frac{1}{4}, \quad \langle \delta \hat{y}_b^2 \rangle(0) = \frac{1}{4}, \quad (17c)$$

$$\langle \delta \hat{x}_b \delta \hat{x}_i \rangle(0) = 0, \quad \langle \delta \hat{y}_b \delta \hat{y}_i \rangle(0) = 0, \quad (17d)$$

ここでは、入力されるコヒーレント状態のポンプパルスのq成分変位として $\beta := \langle \hat{q}_b \rangle(0)$ を導入した。すなわち、その振幅は $\beta/\sqrt{2}$ 、光子数の期待値は $\beta^2/2$ である。式(11)の写像を実装するには、これらの初期条件から、パルスが結晶内を通過する時間 τ_{nl} だけ、運動方程式(15)、(16)を積分する。式(10)の非線形結合率 ϵ と伝搬時間 τ_{nl} は、このモデルでは無次元積 $\epsilon \tau_{nl}$ としてのみ現れる。

この常微分方程式系は8個の実数値の動変数からなり、効率よく数値的に解ける。このため、導出では直交位相演算子 $\hat{x}, \hat{y}$ を選んだ。平均と共分散が複素数となるモード演算子 $\hat{a}, \hat{a}^\dagger$ を使うと、8本の複素数値の常微分方程式が必要になるためである。

C. MFB-CIMの離散時間動力学モデル

MFB-CIMのモデル化に必要な構成要素と変換をすべて説明したので、次にそのダイナミクスを生成する具体的な反復手順を示す。第iパルスが第k周回を始める直前の状態を $\rho(\mu^{(i)}(k), \Sigma^{(i)}(k))$ とする。この時点の実時間は $(kN + i)/f_{rep}$ であり、 $1/f_{rep}$ はパルスの繰り返し間隔である。この定義による「状態」 $\otimes_{i=1}^N \rho(\mu^{(i)}(k), \Sigma^{(i)}(k))$ は、厳密には異なる時刻の信号パルス状態を組み合わせている。 $i = N$ のパルスが前の周回を終える前に、 $i = 1$ のパルスは次の周回に入り、すでにいくつかの光学素子と相互作用している可能性があるためである。それでも各パルスの発展はLOCCなので、この微妙な点は大きな問題にはならない。

第i信号パルスの状態を $\rho(\mu^{(i)}(k), \Sigma^{(i)}(k))$ から $\rho(\mu^{(i)}(k+1), \Sigma^{(i)}(k+1))$ へ発展させるには、以下の操作を反復する。

1. 入射端面損失：入射端面での損失は、次の操作でモデル化できる。

$$\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \mapsto \text{tr}_c \left(\mathcal{B}_{r_{loss}} \left[\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \otimes \hat{\rho}(0^{(c)}, \Sigma_0^{(c)}) \right] \right), \quad (18)$$

Bは式(5)のビームスプリッタ写像、 r_{loss}^2 はその端面でのパワー損失である。物理的には、

ically, c represents a vacuum mode, which mixes with the signal pulse and is then traced out.

2. *Crystal propagation*: Following (11), the crystal propagation is described by a Gaussian map producing a joint correlated signal-pump state, followed by a partial trace of the pump mode:

$$\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \mapsto \text{tr}_b \left[\chi \left(\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \otimes \hat{\rho}(\beta^{(b)}, \Sigma_0^{(b)}) \right) \right], \quad (19)$$

where the b mode is a displaced coherent state with mean $\beta^{(b)} := (\beta, 0)$, and the map χ is obtained by solving the nonlinear Gaussian EOMs (15) and (16) with initial conditions (17). As described in Sec. IIB, these EOMs involve the nonlinear interaction strength $\epsilon\tau_{\text{nl}}$ due to the action of the crystal Hamiltonian (10), as well as the initial pump displacement $\beta := \langle \hat{q}_b \rangle(0)$.

3. *Output facet loss*: This step is exactly the same as for the input facet loss. Assuming we can lump the total system losses in a symmetric way between input and output losses around the crystal, we can again apply (18).
4. *Outcoupling and homodyne measurement*: The homodyne measurement consists of two steps. First, a part of the internal signal pulse is outcoupled, which can be described by the map

$$\begin{aligned} \hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) &\mapsto \hat{\rho}(\mu^{(i,h)}, \Sigma^{(i,h)}) \\ &:= \mathcal{B}_{r_{\text{out}}} \left[\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \otimes \hat{\rho}(0^{(h)}, \Sigma_0^{(h)}) \right], \end{aligned} \quad (20a)$$

where r_{out}^2 is the power outcoupling. This takes a probe external mode h initialized in the vacuum state and mixes it with the signal pulse at the outcoupler to produce a correlated state of the internal cavity mode and the external outcoupled mode. The next step is to apply a homodyne measurement on the outcoupled mode, which produces a measurement result $w_i(k)$ for the i th signal pulse at this roundtrip index k according to (7). This indirect measurement of the internal signal pulse projects its state according to the map

$$\begin{aligned} \hat{\rho}(\mu^{(i,h)}, \Sigma^{(i,h)}) &\mapsto \mathcal{M}_h \left[\hat{\rho}(\mu^{(i,h)}, \Sigma^{(i,h)}) \right] \\ &= \hat{\rho}(\mu_{w_i}^{(i)}, \Sigma_{w_i}^{(i)}), \end{aligned} \quad (20b)$$

where $\mathcal{M}$ is the conditional homodyne map (9), with the mean and variances computed via (8).

5. *Measurement-based feedback injection*: We finally apply displacements to the signal pulses based on the feedback signal computed by the FPGA for implementing the Ising couplings. Let the feedback terms be given by

$$v_i(k) = J_0(k) \sum_{j=1}^N J_{ij} w_j(k), \quad (21a)$$

where $w_i(k)$ are the measurement results from the homodyne detection in this roundtrip, and $J_0(k)$ is a feedback gain parameter, which may generally depend on the roundtrip index k (i.e., time). We now displace the pulse amplitudes according to

$$\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \mapsto \mathcal{V}_{v_i} \left[\hat{\rho}(\mu^{(i)}, \Sigma^{(i)}) \right], \quad (21b)$$

where $\mathcal{V}$ is the displacement operation given by (6).

The above steps, after being applied to each pulse $i = 1, \dots, N$, completes one roundtrip through the CIM cavity. Note that the exact order in which we apply the above operations depends on the details of how the cavity is laid out and the relative time-of-flight between optical components, and our choice above is to some extent arbitrary. Nevertheless, generic features such as steady-state behavior should be robust against the exact choice of ordering, and if the precise transient behavior is desired (which can be important for very low-finesse operation), one can rearrange the procedure above to more accurately model the specific cavity layout.

D. Reduction to continuous-time Gaussian models

In this subsection, we briefly summarize how our discrete-time model can be reduced to continuous-time models more conventionally used in studies of optical CIMs in the “high-finesse”. A complete derivation of this correspondence is presented in Appendix A, and we only summarize the key ideas and results here.

In the high-finesse limit, each discrete operation only implements an infinitesimal change $\hat{\rho} \mapsto \hat{\rho} + d\hat{\rho}$ to the state $\hat{\rho}$ and, as in the Trotterization of quantum dynamics [42, 43], the exact order in which the operations are composed within one roundtrip becomes unimportant, allowing us to analyze the operations in Sec. IIC independently within one roundtrip.

We introduce a parameter δ such that $\delta \rightarrow 0$ formally defines the high-finesse limit. We begin with the assumption that the MFB-CIM roundtrip time (as measured by a wall clock) is $\Delta t = N/f_{\text{rep}} \sim \delta$. We then also assume that the model parameters in Sec. IIC scale as follows:

$$r_{\text{loss}}^2 \sim r_{\text{out}}^2 \sim \beta^2 \sim (\epsilon\tau_{\text{nl}})^2 \sim J_0^2 \sim \delta. \quad (22)$$

As necessitated by working in the Gaussian regime, we also need to assume, for any fixed δ , $(\epsilon\tau_{\text{nl}})^2 \ll r_{\text{loss}}^2 + r_{\text{out}}^2$.

In Appendix A, we analyze each step of the discrete-map iteration from Sec. IIC by expanding their effects on the Gaussian means and variances $\langle \hat{q}_i \rangle$ and $\langle \delta \hat{q}_i^2 \rangle$ up to first order in δ . Notably, the crystal propagation step can be similarly treated by using Picard iteration to integrate (15) and (16) to first order in δ , thus capturing the effects of parametric gain and pump depletion. After going through one entire roundtrip, we end up with updated means and variances with corrections up to first order in δ . Denoting the updated state variables with

a prime, the discrete-time map dynamics can then be connected to continuous-time derivatives via

$$\lim_{\Delta t \rightarrow 0} \frac{\langle \hat{q}_i \rangle' - \langle \hat{q}_i \rangle}{\Delta t} := \frac{d\langle \hat{q}_i \rangle}{dt}, \quad (23a)$$

$$\lim_{\Delta t \rightarrow 0} \frac{\langle \delta \hat{q}_i^2 \rangle' - \langle \delta \hat{q}_i^2 \rangle}{\Delta t} := \frac{d\langle \delta \hat{q}_i^2 \rangle}{dt}. \quad (23b)$$

The continuous-time differential equations of motion have the explicit form

$$\begin{aligned} \frac{d\langle \hat{q}_i \rangle}{dt} = & (p - \kappa - \gamma) \langle \hat{q}_i \rangle - \frac{g}{2} \langle \hat{q}_i \rangle^3 + \lambda \sum_{j=1}^N J_{ij} \langle \hat{q}_j \rangle \\ & + 2\sqrt{\kappa} \left(\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right) \xi_i + \frac{\lambda}{2\sqrt{\kappa}} \sum_{j=1}^N J_{ij} \xi_j \end{aligned} \quad (24a)$$

$$\begin{aligned} \frac{d\langle \delta \hat{q}_i^2 \rangle}{dt} = & 2p \langle \delta \hat{q}_i^2 \rangle - 2(\gamma + \kappa) \left(\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right) \\ & - 4\kappa \left(\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right)^2 - 2g \langle \hat{q}_i \rangle^2 \left(\frac{3}{2} \langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right). \end{aligned} \quad (24b)$$

These equations are specified by the rates γ (the intrinsic loss rate), κ (the outcoupling rate), p (the pump rate), g (the nonlinear rate), and λ (the feedback Ising-coupling rate), together with a set of white-noise processes ξ_i obeying $\langle \xi_i(t), \xi_j(t') \rangle = \delta_{ij} \delta(t - t')$. In the limit $\Delta t \rightarrow 0$, they can be expressed in terms of discrete-time parameters as

$$\gamma := \frac{r_{\text{loss}}^2}{\Delta t}, \quad \kappa := \frac{r_{\text{out}}^2}{2\Delta t}, \quad (25a)$$

$$p := \frac{\beta \epsilon \tau_{\text{nl}}}{\sqrt{2}\Delta t}, \quad g := \frac{(\epsilon \tau_{\text{nl}})^2}{4\Delta t}, \quad (25b)$$

$$\lambda := \frac{J_0 r_{\text{out}}}{\Delta t}, \quad \xi_i := \frac{z_i}{\sqrt{\Delta t}}. \quad (25c)$$

As discussed in Appendix A, continuous-time EOMs of the form (24) have recently been shown to arise from quantum-optical master equations under appropriate Gaussian-state assumptions [34, 35], where the rates (25) are the basic parameters in those models. Thus, while our model captures dynamics in the MFB-CIM beyond the high-finesse limit, it also reproduces the diffusive dynamics predicted by traditional quantum-optical models for the MFB-CIM in the appropriate limits. As such, Appendix A may serve as a useful reference for readers interested in further exploring the relationship between continuous-time and discrete-time models of the MFB-CIM.

III. NUMERICAL RESULTS

In this section, we present and discuss small-scale numerical simulations of the discrete-time Gaussian model presented in Sec. II. We first show some representative trajectories of the model dynamics and define a suitable

metric for sampling performance. We explore how sampling performance for a single problem instance depends on various model parameters, such as feedback gain or cavity finesse, and we verify that the sampling behavior is consistent across many different problem instances at small scale. Finally, we try to gain insight into the MFB-CIM dynamics by numerically studying a handful of operational modifications to the conventional MFB-CIM, such as the use of negative parametric gain, the removal of optical nonlinearity, and the replacement of quantum noise with classical noise.

A. Model parameters

For our numerical results, it is useful to define a new set of parameters, which scale the model more conveniently by keeping certain qualitative features of the dynamics constant. The dynamics of an uncoupled classical DOPO are critically determined by three parameters: (1) the cavity-photon $1/e^2$ -decay time in the absence of pumping; (2) the pump parameter $r = \beta/\beta_{\text{th}}$ giving the ratio between the pump field β over its threshold value β_{th} ; and (3) the saturation photon number n_{sat} . We express each of these quantities in terms of the model parameters used in Sec. II. For convenience, let us define

$$R_{\text{out}} := r_{\text{out}}^2, \quad (26a)$$

$$R_{\text{loss}} := 1 - (1 - r_{\text{loss}}^2)^2, \quad (26b)$$

where the latter quantity represents the total fraction of power lost through *both* facets.

First, in the absence of pumping or nonlinearity, the number of roundtrips T_{decay} required for the photon number to attenuate by a factor of $1/e^2$ due to linear loss and outcoupling is simply given by

$$1/T_{\text{decay}} := -\log \left[(1 - R_{\text{out}})^{1/2} (1 - R_{\text{loss}})^{1/2} \right]. \quad (27)$$

In addition, because T_{decay} captures the effect of r_{loss} and r_{out} together, it is also convenient to define an “escape efficiency” parameter

$$\eta_{\text{esc}} := \frac{R_{\text{out}}}{1 - (1 - R_{\text{out}})(1 - R_{\text{loss}})}, \quad (28)$$

which captures the relative amount of (power) attenuation due to outcoupling as opposed to loss.

The threshold pump field is taken to be the value of β (i.e., the q -quadrature displacement of the input pump pulse) such that the exponential gain experienced by a small-signal input to the crystal (i.e., a signal pulse with vanishing amplitude) exactly balances the attenuation due to linear loss and outcoupling. The pump parameter is then simply the pump field divided by this threshold value β_{th} . We therefore define

$$r := \frac{\beta}{\beta_{\text{th}}}, \quad \text{where} \quad \beta_{\text{th}} := \frac{\sqrt{2}}{\epsilon \tau_{\text{nl}}} \frac{1}{T_{\text{decay}}}. \quad (29)$$

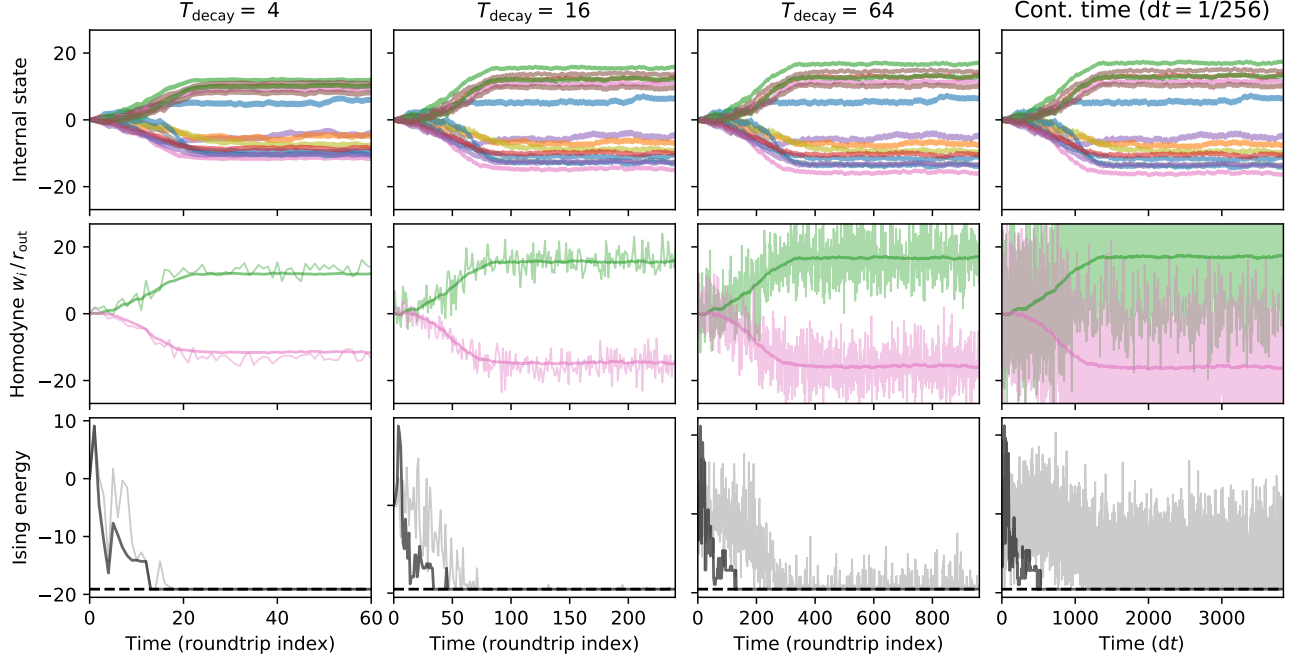


FIG. 3. Representative trajectories of the discrete-time Gaussian-state MFB-CIM model for the $N = 16$ problem instance shown in Fig. 2. Each of the first three columns represents a different cavity decay time T_{decay} , while the rightmost column depicts the continuous-time limit sampled at $dt = 1/256$; all trajectories are based on the same underlying noise process but sampled at different rates. The N traces shown in the first row represent the Gaussian states of the N intracavity signal pulses: the center of each trace gives the mean quadrature amplitude $\langle \hat{q}_i \rangle$, while the half-thickness of each trace is given by the root-variance $\langle \delta \hat{q}_i^2 \rangle^{1/2}$. The second row shows two of the homodyne records w_i (specifically $i = 2, 3$) divided by the outcoupling ratio r_{out} ; the corresponding intracavity mean $\langle \hat{q}_i \rangle$ is reproduced from the first row as a guide for the eye. In the last row, the dark line shows the Ising energy $-\sum_j J_{ij} \text{sgn}(\langle \hat{q}_i \rangle) \text{sgn}(\langle \hat{q}_j \rangle)$ based on the intracavity mean, while the light-grey line shows the Ising energy $-\sum_j J_{ij} \text{sgn}(w_i) \text{sgn}(w_j)$ based on the homodyne record (black dashed line shows the ground energy). In this simulation, the other model parameters are held fixed at $n_{\text{sat}} = 200$, $\eta_{\text{esc}} = 0.5$, $r = 0.9$, and $\alpha = 5$.

The saturation photon number is the steady-state photon number at $r = 2$, considering the effects of loss and outcoupling, parametric gain, and nonlinear gain saturation. Because this feature involves the nonlinear terms of the crystal EOMs at finite signal amplitude, the exact value of the saturation photon number can depend on cavity layout for low-finesse cavities. In the high-finesse limit, however, it can be shown to be

$$n_{\text{sat}} = \frac{8}{(\epsilon\tau_{\text{nl}})^2} \frac{1}{T_{\text{decay}}}. \quad (30)$$

When the roundtrip attenuation is moderately low (~ 0.4 in power), then $\epsilon\tau_{\text{nl}} \ll 1$, and we can take for convenience the above equation to *define* the parameter n_{sat} , so that for a fixed T_{decay} , specifying n_{sat} determines $\epsilon\tau_{\text{nl}}$, which then fixes β_{th} . When the roundtrip attenuation is large, however, it may be the case that a given n_{sat} corresponds to $\epsilon\tau_{\text{nl}} \not\ll 1$, which is inconsistent with a Gaussian-state approximation for the crystal propagation. To handle these cases as well, we specify

$$(\epsilon\tau_{\text{nl}})^2 = \min\left(\frac{8}{n_{\text{sat}} T_{\text{decay}}}, 10^{-2}\right), \quad (31)$$

where 10^{-2} is taken as an appropriate maximal value to respect the Gaussian-state approximation. In this latter case, (29) and (30) are replaced with $\beta_{\text{th}} = (10\sqrt{2})/T_{\text{decay}}$ and $n_{\text{sat}} = 800/T_{\text{decay}}$.

In summary, we henceforth parametrize our system using the values T_{decay} , η_{esc} , r , and n_{sat} . We use (27) and (28) to determine r_{loss} and r_{out} , and we use (31) to determine $\epsilon\tau_{\text{nl}}$. This procedure sets β_{th} via (29), which also gives us β given r .

Finally, with regards to the feedback control, we note that for $r < 1$, the feedback gain J_0 needed for the system to go above threshold due to feedback gain scales with $\sqrt{T_{\text{decay}}}$ but also with the Ising matrix entries J_{ij} . To this end, we define a feedback gain parameter

$$\alpha := -J_0 \sqrt{T_{\text{decay}}} \left(\sum_{i \neq j} |J_{ij}| \right)^{1/2}, \quad (32)$$

where the negative sign is chosen since we usually use $J_0 < 0$ in order for the feedback to enforce minimization of the Ising energy; thus α is positive in these cases.

In Fig. 3, we illustrate some representative dynamics of the MFB-CIM running on the $N = 16$ problem instance shown in Fig. 2. As a way of making the discussion in Sec. IID more concrete, we note in particular how these

trajectories change as a function of T_{decay} while all other model parameters are held constant. By running the simulation for $15T_{\text{decay}}$ roundtrips in all cases, we see there is a qualitative difference in going from a low-finesse system ($T_{\text{decay}} = 4$), to an intermediate-finesse one ($T_{\text{decay}} = 16$), to a high-finesse one ($T_{\text{decay}} = 64$), but the dynamics eventually converge to the continuous-time trajectory in the high-finesse limit, as depicted in the rightmost column. As expected, the homodyne record w_i (and hence the measured Ising energy $-\sum_{j=1}^N J_{ij} \text{sgn } w_i \text{sgn } w_j$) becomes increasingly noisy as the finesse increases because the outcoupling ratio $r_{\text{out}} \sim 1/\sqrt{T_{\text{decay}}}$, providing less information about the internal state in any given shot of the measurement and requiring more roundtrips to obtain the same amount of information produced by a single measurement shot in a lower-finesse system. In addition, if we assume the wall-clock roundtrip time is fixed (corresponding to the time between successive points in the discrete-time model, or to dt in the continuous-time model), it follows that the low-finesse system takes a shorter wall-clock time to reach steady state (i.e., ~ 40 roundtrips at $T_{\text{decay}} = 4$ vs ~ 640 roundtrips at $T_{\text{decay}} = 64$), all else being equal. This scaling plays an important role in the efficiency of the system and the overall time-to-sample as we analyze next.

B. Ising sampling in Gaussian MFB-CIMs

As evident from Fig. 3, the dynamics of the MFB-CIM drive the system towards states encoding low-energy spin configurations of the Ising problem. At the same time, the particular configurations found by the MFB-CIM are stochastic. We therefore expect that, at least in certain regimes of operation, the MFB-CIM can be used to *stochastically sample* different spin configurations, simply by running the system under the injection of measurement noise. Each “run” of the MFB-CIM would consist of a homodyne record like the ones shown in Fig. 3, which takes $\mathcal{O}(T_{\text{decay}})$ roundtrips to collect and would yield one or more samples of low-energy Ising spin configurations after an initial transient period; repeated runs (i.e., passing through threshold again) could also generate new samples. The sampling efficiency is thus characterized by the likelihood of a given trajectory to yield at least one sample of interest and also how quickly it can do so, while sampling fairness depends on how uniformly such configurations are distributed.

Figure 4 illustrates this procedure for the $N = 16$ problem instance shown in Fig. 2. In Fig. 4(a), the number of trajectories where each spin configuration appears at least once is recorded. Over the course of 1000 trajectories, we easily obtain multiple samples of every spin configuration of interest, indicating relatively fair sampling at least for this instance. Nevertheless, there are systematic biases in the sampling: namely, the spin configurations in a given energy level are not necessarily uniformly sampled. These biases are problem-dependent in

general but also depend on the model parameters chosen for the sampling process.

To fully quantify the sampling efficiency, however, it is not enough to simply count trajectories in which each spin configuration appears, since certain configurations may systematically appear later than other configurations within any given trajectory. These differences in sampling time are illustrated in Fig. 4(b), where, for each spin configuration considered in Fig. 4(a) (up to an overall sign flip), we show a (vertical) histogram of the *first time* that configuration appeared in each trajectory, if

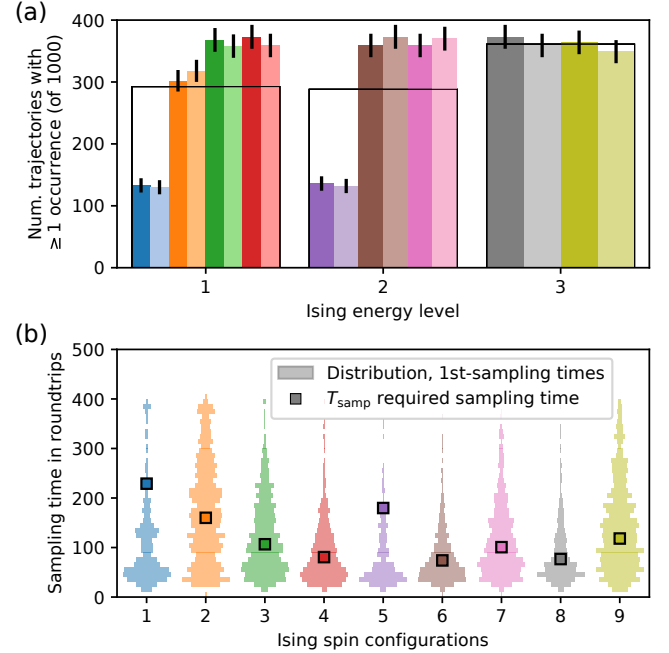


FIG. 4. Sampling low-energy spin configurations of the $N = 16$ Ising problem shown in Fig. 2 using an MFB-CIM. (a) Bar graph showing the number of trajectories with at least one occurrence of each spin configuration, each represented by a colored bar (adjacent bars of the same color but different lightness represent configurations differing only in an overall sign flip). Configurations are grouped together by Ising energy, with labels 1, 2, and 3 for the ground, first-excited, and second-excited energy levels, respectively. The solid-line rectangle around each group represents the expected histogram if sampling within that energy level were perfectly fair. Error bars indicate simulation uncertainty of the counts. (b) Resolving the time along a trajectory at which a sample first appears. The vertical histograms show, for the set of realized samples, the distribution of time (in roundtrips) at which the first sample appeared (i.e., the first-sampling time). Each spin configuration is colored in accordance with (a); however, occurrences of configurations differing only in an overall sign flip have been combined. Square markers show the required sampling time based on T_{samp} as defined by (33), a metric taking into account both the first-sampling time distribution as well as the frequency of occurrences. For this simulation, $r = 0.8$, $\alpha = 5$, $T_{\text{decay}} = 4$, $\eta_{\text{esc}} = 0.2$, and $n_{\text{sat}} = 200$; $N_{\text{traj}} = 1000$ trajectories are simulated for $100T_{\text{decay}}$ roundtrips each.

it appeared at all. There is an initial transient period ($\sim T_{\text{decay}}$ roundtrips) in which low-energy samples cannot be generated, and generally most of the distributions are peaked within a few decay times of the transient. However, the exact distribution of this first-sampling time differs across spin configurations, with some featuring sharper peaks and others having longer tails. As a result, a configuration appearing less often on a per-trajectory basis may nevertheless be efficient to sample if it tends to appear earlier.

We can define a “required sampling time” metric in order to take into account these effects, including the biases in overall counts, the transient-time costs, and the variation in the first-sampling-time distributions. Let us suppose we have collected an ensemble of homodyne records $w_i^{(\ell)}(k)$, where $1 \leq \ell \leq N_{\text{traj}}$ denotes different trajectories, $1 \leq i \leq N$ denotes the DOPO or spin index, and $k \geq 1$ indexes the number of roundtrips elapsed. Suppose further we are interested in sampling a particular Ising spin configuration $\sigma := (\sigma_1, \dots, \sigma_N)$. Then we define the first-sampling time of σ in trajectory ℓ as

$$T_{\text{samp}}^{(\ell)}(\sigma) := \min_{k \geq 0} \left\{ k : \forall i \leq N, \text{sgn}(w_i^{(\ell)}(k)) = \sigma_i \right\}, \quad (33a)$$

where we take $\min \emptyset = \infty$ by convention for trajectories that produce no samples of σ . Then given a sufficiently large number of trajectories N_{traj} , each simulated for a sufficiently long time, an estimate for the required number of roundtrips to sample σ is $T_{\text{samp}}(\sigma)$, where

$$\frac{1}{T_{\text{samp}}(\sigma)} := \frac{1}{N_{\text{traj}}} \sum_{\ell=1}^{N_{\text{traj}}} \frac{1}{T_{\text{samp}}^{(\ell)}(\sigma)}. \quad (33b)$$

Under this metric, a spin configuration that is realized less often (so $1/T_{\text{samp}}^{(\ell)} = 0$ for more values of ℓ) will have a larger T_{samp} , as will a spin configuration that takes longer to appear (so $T_{\text{samp}}^{(\ell)}$ is finite but large). As a result, $T_{\text{samp}}(\sigma)$ captures, in an *a posteriori* sense, the observed efficiency for sampling the configuration σ .

Having defined this empirical measure of required sampling time, we can turn to how it is affected by the model parameters, especially the feedback gain α and the pump parameter r . Figure 5 shows how, for the 9 configurations considered in Fig. 4, the largest required sampling time T_{samp} varies with α and r . Efficient sampling in the MFB-CIM is relatively robust across a wide range of system parameters. The most critical parameters are indeed the feedback gain and the pump parameter, which show a sharp cutoff in sampling performance near the estimated linear threshold of the MFB-CIM. The required sampling time is lower for systems with a faster cavity decay time (i.e., $T_{\text{decay}} = 4$), which reflects the fact that a low-finesse cavity spends fewer roundtrips in the transient period and can yield low-energy samples more quickly due to larger (i.e., nondiffusive) kicks from the noise in each step. There appears to be a slight advantage to using a system with lower escape efficiency (higher background

losses), which may be related to the fact that background loss affects the dynamical correlation between the pulse amplitudes differently from the noise due to measurement outcoupling [29, 35].

One particularly important aspect of the sampling behavior in the MFB-CIM is that the required sampling time scales with the finesse of the system as measured by T_{decay} . To check whether this scaling is robust with respect to the choice of problem instance, we consider a set of integer-valued Sherrington-Kirkpatrick spin-glass Ising problems with range 1 (SK1), which is equivalent to a set of MAX-CUT problems with binary-signed edge weights. In Fig. 6, we show the distribution of $T_{\text{samp}}(\sigma)$ over 50 SK1 $N = 16$ problem instances, where, for concreteness, σ is chosen to be the first lexicographic spin configuration that gives the ground energy of the problem. We see that although there is a spread in the required sampling time across problem instances, the distributions are characterized by means and medians, which show a clear monotonic decrease with decreasing decay time. Interestingly, this scaling persists to very low decay times on the order of $T_{\text{decay}} \sim 1$, which is well outside the validity of any high-finesse or continuous-time model. In fact, for this problem size, performance only saturates and degrades at $T_{\text{decay}} \approx 0.2$, which, for this parameter set, is the point at which the roundtrip attenuation begins to exponentially approach unity (i.e., $R_{\text{loss}} \sim 1 - e^{-c/T_{\text{decay}}}$) as a function of T_{decay} . At this point, the sensitivity of the system to

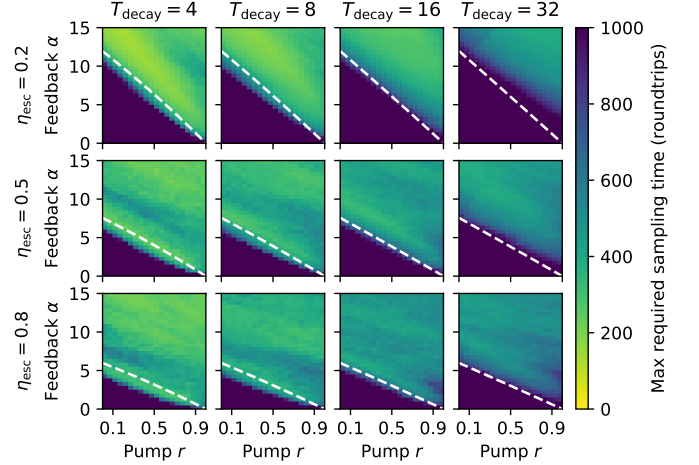


FIG. 5. Required sampling time T_{samp} in roundtrips as a function of model parameters for the problem instance and spin configurations considered in Fig. 4. For each choice of model parameters, we run $N_{\text{traj}} = 1000$ trajectories for $100T_{\text{decay}}$ roundtrips and compute $T_{\text{samp}}(\sigma)$ according to (33) for each spin configuration σ , taking the maximum over the 9 configurations. As in Fig. 4(b), we combine samples from configurations differing only in an overall sign flip. The dashed-white line indicates the threshold of the system, defined to be the boundary in α and r , beyond which there exists a nonzero-amplitude fixed point of the system dynamics at steady state. In all these simulations, we fix $n_{\text{sat}} = 200$.

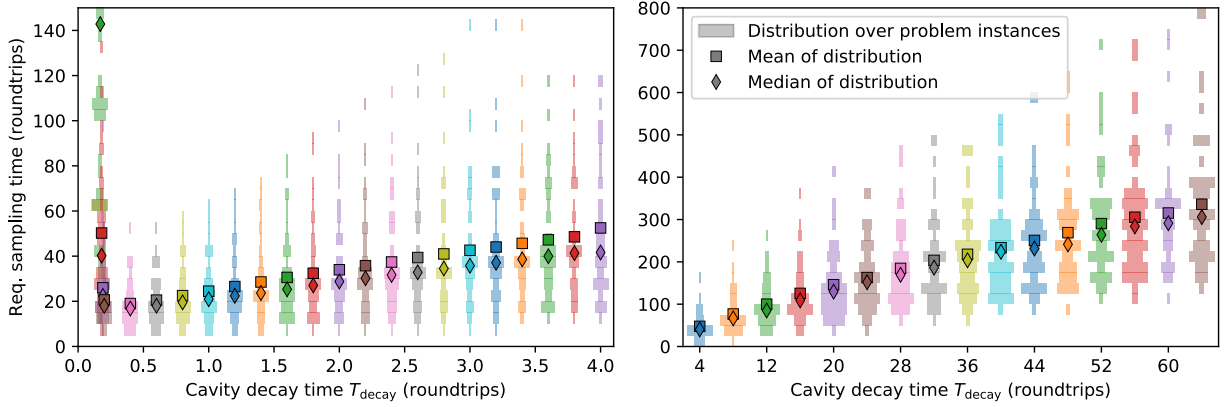


FIG. 6. Time $T_{\text{samp}}(\sigma)$ required to sample a specified Ising ground-state configuration σ as a function of the cavity decay time T_{decay} for a set of 50 $N = 16$ SK1 problems. For concreteness, we choose σ as the first lexicographic ground-state configuration each problem (up to an overall sign flip). Each vertical histogram shows the distribution of $T_{\text{samp}}(\sigma)$ as defined by (33) over the different problem instances. Diamond markers indicate the median of the distribution while squares indicate the mean. Note that when the required sampling time becomes large, the mean may not be defined due to some problem instances requiring more trajectories to sample than were performed. For these simulations, we take $r = 0.8$, $\alpha = 4$, $\eta_{\text{esc}} = 0.2$, and $n_{\text{sat}} = 200$; we run $N_{\text{traj}} = 1000$ trajectories for $100T_{\text{decay}}$ roundtrips.

system parameters precludes any additional significant gains in reducing the sampling time. The fact that the sampling performance continues to improve into the low-finesse regime is a key motivation for the development of our discrete-time Gaussian-state model.

C. Sampling in alternative models of MFB-CIM

Although our focus thus far has been on developing a general model for the MFB-CIM in the Gaussian-state approximation and studying the dynamical role of quantum noise in its conventional operation (with parametric gain, homodyne measurement/feedback, measurement backaction, and gain saturation), it is also useful to consider alternative models or modes of operation, which may be conceptually simpler or easier to implement experimentally. Of particular interest is to relate our quantum-based model to established classical analogs or formulations of CIMs, such as those based on coherent-state feedback networks without nonlinearity [37], or those based on deterministic nonlinear dynamics (with no quantum noise and only a random initial condition), which have proven to be fruitful models in which to study the roles of feedback and nonlinearity for CIM combinatorial optimization [44–46].

In this subsection, we consider three cases:

1. *MFB-CIM with zero or negative parametric gain:* Conventionally, the MFB-CIM is operated with parametric gain, i.e., the pump parameter $r > 0$. However, we can also explore its sampling performance for $r \leq 0$, with the case of $r = 0$ being especially experimentally interesting as it does not require a pump source. Such modifications are

straightforward within our general Gaussian model, so we can directly compare these cases against the conventional $r > 0$ case while keeping gain saturation, quantum noise, and so on fixed.

2. *MFB-CIM without nonlinear crystal:* We can also go one step further and consider a MFB-CIM without any parametric interaction (i.e., no optical nonlinearity) by setting $\epsilon\tau_{\text{nl}} = 0$, resulting in a “coherent-state” MFB-CIM, where the internal field is only excited through external coherent-state injection. This model has previously been studied in Ref. [37] in the context of combinatorial optimization (also via a discrete-time formulation), whereas we investigate here its potential for sampling. Since the resulting system has linear dynamics, the Gaussian formalism applies exactly and is an efficient representation of the quantum state throughout the dynamics.
3. *Mean-field MFB-CIM with injected measurement noise:* A common approach to studying open-dissipative optical systems with weak single-photon nonlinearities is to neglect quantum noise altogether by taking a mean-field or classical limit, resulting in deterministic c-number EOMs. We describe how such a limit can be taken for our Gaussian MFB-CIM model, producing not only the usual continuous-time mean-field models for the MFB-CIM [44, 45] but also a discrete-time mean-field model similar to that of Ref. [36] as well. However, to study sampling performance in this limit, we need an alternative noise source in the mean-field model. For this purpose, we supplement the model by injecting fixed-variance Gaussian-distributed noise (limiting to white noise at the

continuous-time limit) in the measurement-and-feedback step [47]; such an extrinsic noise source can correspond, for example, to classical Johnson noise in the detector or to a random signal intentionally generated by the FPGA circuit (e.g., via a pseudorandom number generator).

In Fig. 7(a), we show the maximum required sampling time as a function of the feedback gain over a range of pump parameters r , including $r \leq 0$, for the previously considered case of $T_{\text{decay}} = 4$ and $\eta_{\text{esc}} = 0.2$ from Fig. 5. (For $r \geq 0$, these lines are simply vertical slices of the upper-left panel of Fig. 5.) We see that, surprisingly, performance is quite comparable over a wide range of pump parameters with negative r , corresponding to parametric *deamplification*. In this regime, the system even gives slightly better performance at the expense of requiring higher feedback gain to overcome the deamplification. Generally, for sufficiently large feedback gain, there is always a robust region over which acceptable sampling performance is obtained, but for $r > 0$, there is also a “sweet spot” at lower feedback gain where the stochastic noise due to antisqueezing can allow for efficient sampling with lower feedback gain. In Sec. IV we explore how these two operational modes, $r > 0$ and $r < 0$, scale to larger problem instances.

Next, we investigate the second model where the nonlinear crystal is removed from the MFB-CIM. We set $\epsilon\tau_{\text{nl}} = 0$ (so $n_{\text{sat}} = \infty$), which eliminates the need to integrate (15) and (16) for the crystal propagation each roundtrip. There is also no longer a pump parameter, leaving us with just the feedback gain parameter α in addition to T_{decay} and η_{esc} . Note that without the nonlinear saturation, the system is unstable once the feedback gain exceeds the roundtrip attenuation due to loss and outcoupling, but because our sampling metric (33) only involves the sign of the homodyne result, the metric is unaffected so long as we terminate the simulation before numerical overflow. In Fig. 7(b), we show the maximum required sampling time for this MFB-CIM model at $N_{\text{decay}} = 4$. We find that the performance also exhibits a certain threshold, which occurs at smaller values of feedback gain compared with the nonlinear MFB-CIM. However, the attained sampling times are greater than that of the nonlinear MFB-CIM by at least a factor of two. This observation suggests that the nonlinear saturation plays an important role in effectively embedding the Ising problem into the dynamics of the MFB-CIM, consistent with the findings of Ref. [46]. In Sec. IV we explore how this model scales out to larger problem instances.

While the former two cases are straightforward to address within our model, the third approach involves taking a mean-field limit, which we can motivate as follows. For simplicity, we illustrate the limit using the continuous-time Gaussian-state EOMs (24), although by using the exact mapping detailed in Sec. IID, the procedure for the discrete-time version can be similarly derived. To take the mean-field limit, we define a rescaled mean-field coordinate $\tilde{q}_i := \sqrt{g/\kappa} \langle \hat{q}_i \rangle$, in which case (24)

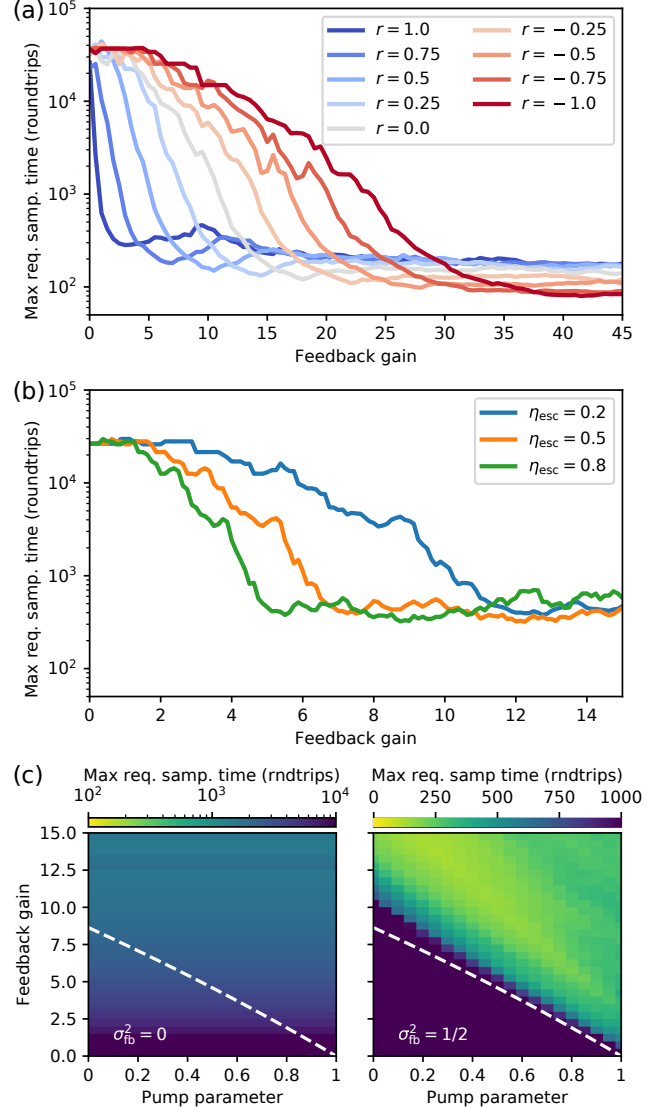


FIG. 7. Required sampling time of various alternative models of MFB-CIM for the problem instance and spin configurations considered in Fig. 4. (a) Required sampling time as a function of the feedback gain parameter α for different values of the pump parameter r , using the full Gaussian model of the MFB-CIM. (b) Required sampling time as a function of α , and with various values of escape efficiency η_{esc} , for the MFB-CIM with no crystal ($\epsilon\tau_{\text{nl}} = r = 0$) and only linear dynamics. (c) Required sampling time as a function of both r and α for the mean-field (i.e., no quantum noise) MFB-CIM, with added classical feedback noise with variance $\sigma_{\text{fb}}^2 = 0$ (left, i.e., no noise) and $\sigma_{\text{fb}}^2 = 1/2$ (right). The dashed-white line indicates the linear threshold of the system, as in Fig. 5. In (a) and (b), we fix $n_{\text{sat}} = 200$ and $\eta_{\text{esc}} = 0.2$; the cavity decay time is set to $T_{\text{decay}} = 4$ throughout.

can be written as

$$\frac{d\tilde{q}_i}{dt} = (p - \kappa - \gamma)\tilde{q}_i - \frac{\kappa}{2}\tilde{q}_i^3 + \lambda \sum_{j=1}^N J_{ij}\tilde{q}_j \quad (34a)$$

$$+ \sqrt{g} \left[2 \left(\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right) \xi_i + \frac{\lambda}{\kappa} \sum_{j=1}^N J_{ij} \xi_j \right],$$

$$\frac{d\langle \delta \hat{q}_i^2 \rangle}{dt} = 2p \langle \delta \hat{q}_i^2 \rangle - 2(\gamma + \kappa) \left(\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right) \quad (34b)$$

$$- 4\kappa \left(\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right)^2 - 2\kappa \tilde{q}_i^2 \left(\frac{3}{2} \langle \delta \hat{q}_i^2 \rangle - \frac{1}{2} \right).$$

We now consider the limit of small single-photon nonlinearities where g is very small. As long as $\tilde{q}_i$ is finite, the dynamics of $\langle \delta \hat{q}_i^2 \rangle$ are bounded, so the noise terms in the second line of (34a) scale overall as $\sqrt{g}$, thus becoming negligible compared to the other terms of (34a) in the limit $g \ll \kappa, p, \gamma, \lambda$. It also follows that $\langle \hat{q}_i \rangle \gg \langle \delta \hat{q}_i^2 \rangle$, so that we can neglect the quantum fluctuations, upon which the dynamics are completely characterized by $\tilde{q}_i$, i.e., the EOMs are simplified to

$$\frac{d\tilde{q}_i}{dt} = (p - \kappa - \gamma)\tilde{q}_i - \frac{\kappa}{2}\tilde{q}_i^3 + \lambda \sum_{j=1}^N J_{ij}\tilde{q}_j. \quad (35)$$

The numerical simulations we use to study sampling performance in this limit uses the discrete-time version of the above limit, which involve the same arguments as above. When studying mean-field dynamics, it is standard procedure to introduce a small, random initial condition to avoid unstable fixed points of the dynamics, so we also adopt this convention by setting $\tilde{q}_i(0) = \sigma_i \tilde{q}_0$, where we fix $\tilde{q}_0 := 10^{-3}$ and σ_i is uniformly sampled from ± 1 . We note the main requirement is that $\tilde{q}_0$ be sufficiently small to avoid undue transients in the mean-field simulations. This can correspond, e.g., to an initial seed amplitude much smaller than those produced by the dynamics we are interested in (or indeed by any other physical effects that can destabilize an unstable fixed point).

Because $\langle \delta \hat{q}_i^2 \rangle / \langle \hat{q}_i \rangle^2 \sim g/\kappa \rightarrow 0$, the fluctuations in the homodyne measurement results $\tilde{w}_i := \sqrt{g/\kappa} w_i$ also become negligible in this limit, and $\tilde{w}_i \rightarrow r_{\text{out}} \tilde{q}_j$. The internal cavity state, represented by simply $\tilde{q}_i$, experiences no backaction (e.g., amplitude shift) upon measurement, and the feedback signal $\tilde{v}_i := J_0 \sum_{j=1}^N J_{ij} \tilde{w}_j \rightarrow r_{\text{out}} J_0 \sum_{j=1}^N J_{ij} \tilde{q}_j$ becomes a deterministic function of the internal state. If we wish to restore stochasticity while still retaining the classical character of the model, we can replace the feedback term (21) with

$$\tilde{v}_i := J_0 \sum_{j=1}^N J_{ij} (\tilde{w}_j + z_j), \quad (36)$$

where $z_j \sim \mathcal{N}(0, \sigma_{\text{fb}})$, representing the injection of classical noise into the feedback signal.

In Fig. 7(c), we show the maximum required sampling time as a function of pump parameter and feedback gain for this mean-field model at $N_{\text{decay}} = 4$ and $\eta_{\text{esc}} = 0.2$.

The left panel shows the performance of the mean-field model with $\sigma_{\text{fb}}^2 = 0$, as is conventionally used to study combinatorial optimization in the mean-field MFB-CIM. We find that this model is significantly less efficient at sampling than the Gaussian-state quantum model. On the other hand, setting $\sigma_{\text{fb}}^2 = 1/2$ in the right panel recovers much of the sampling performance of the Gaussian-state quantum model. This result suggests that efficient sampling in the MFB-CIM, while naturally accessible via quantum noise, can nevertheless be largely emulated by classical noise interacting with weak single-photon nonlinearities. Of course, this comparable performance comes at a cost: whereas $|v_i|^2$ and $|\beta|^2$ represent the approximate number of photons (i.e., quanta of energy) required to operate the feedback and pump terms of the Gaussian MFB-CIM, respectively, these energy costs are scaled by a factor of κ/g into the large-photon-number regime for the mean-field MFB-CIM, and this is before accounting for the energy consumption, if any, associated with generating the classical noise z_i . Thus, despite the promising sampling performance predicted for the noisy mean-field MFB-CIM model, it incurs the cost of energy inefficiency compared to the MFB-CIM sampler driven by quantum noise.

We also remark that both the coherent-state linear model and the mean-field nonlinear model explicitly exclude, each in their own way, the quantum correlations between the internal and outcoupled pulses (i.e., $\langle \hat{q}_i \hat{q}_h \rangle$). Thus these two models do not feature the measurement-induced shifts in the mean and variance reduction of the internal state as described by (8) in the Gaussian model. Further research into the dynamical and operational differences among these models could help further elucidate the role of quantum effects in the mechanics of the CIM.

IV. SCALING ESTIMATES OF SAMPLING PERFORMANCE

In this section, we study the scaling of sampling performance in the discrete-time MFB-CIM with respect to problem size. We investigate the extent to which the observations and results from Sec. III obtained from studying small and particular problem instances can generalize to larger sets of larger problems. To be concrete, we focus on the SK1 problem class introduced previously as it features instances with a large number of ground and first-excited spin configurations, and we evaluate the sampling performance of the MFB-CIM with multiple instances of this problem class at every given problem size. We also numerically study the relationship between sampling performance and the degree of degeneracy, as well as the relative scalings among the various alternative models of the MFB-CIM discussed in Sec. III C.

Here, we employ a more stringent metric than the (previously employed) required sampling time T_{samp} to characterize the sampling performance of the MFB-CIM. Operationally, the previous metric attempts to capture

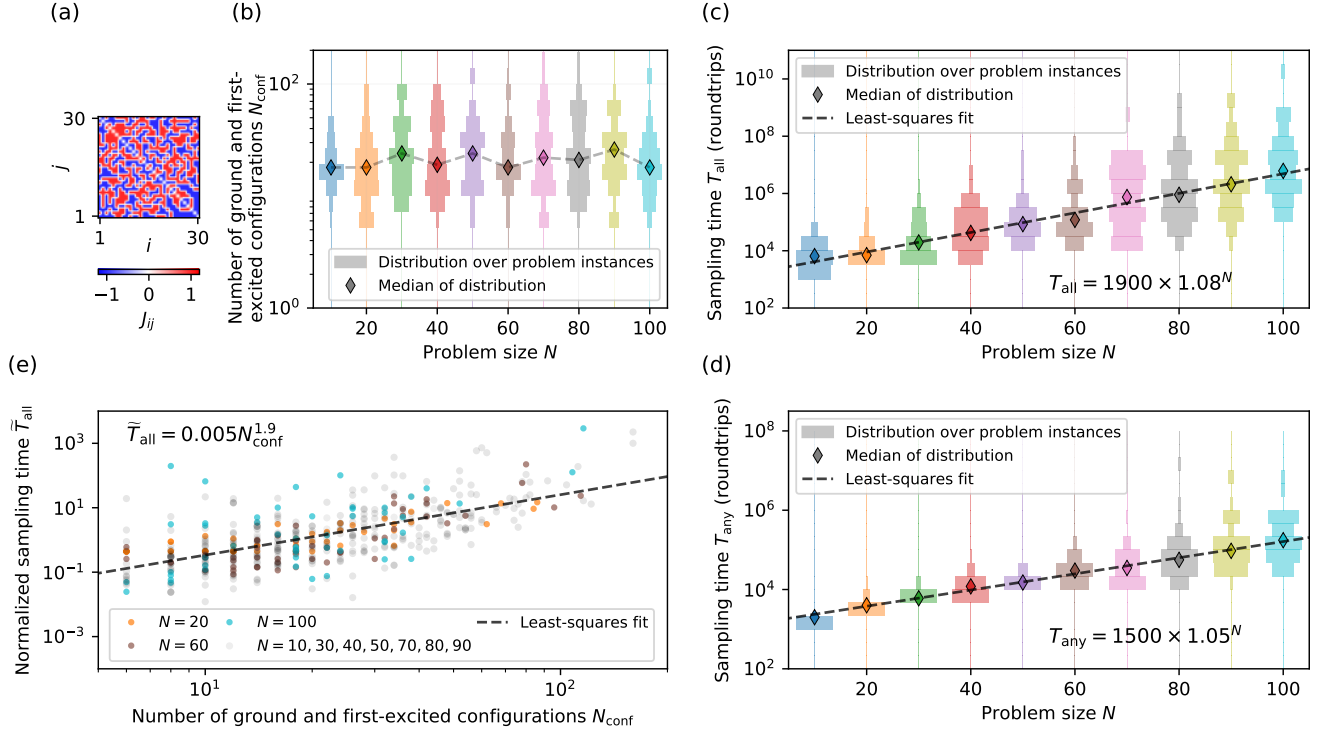


FIG. 8. Scaling of the sampling performance of the MFB-CIM with respect to problem size N , evaluated using 50 SK1 problem instances at each N . (a) A representative Ising problem matrix J_{ij} of the SK1 problem class at $N = 30$. (b) The distribution of the number N_{conf} of ground and first-excited configurations for the SK1 problem class for each N . (c) The distribution of the time T_{all} sufficient to sample all ground and first-excited configurations for each N . (d) The distribution of the time T_{any} sufficient to sample any one of the ground or first-excited configurations for each N . The dashed black lines represent the least-squares fit of the medians of the distributions with respect to N ; the equations of the resultant fits are shown in the respective plots. (e) The normalized time to sample all ground and first-excited configurations as a function of N_{conf} . The normalized sampling time $\bar{T}_{\text{all}}$ is defined as T_{all} divided by the median value of T_{all} , as shown in panel (c), for the respective problem size N . The dashed line represents a least-squares fit on all of the data with respect to N_{conf} ; the equation of the resultant fit is shown in the plot. In these simulations, the MFB-CIM is operated with negative pump parameter; for more details on the model parameters used, see the “Negative pump” row of Table I.

a *necessary* runtime for sampling, which is useful for characterizing the potential computational power of the MFB-CIM but does not prescribe a *sufficient* runtime for sampling that, e.g., can be used in an experimental setting. Thus in this section, we define a sampling time T_{all} given by the number of trajectories taken to sample all ground and first-excited configurations, multiplied by a fixed number of roundtrips T_{sim} (i.e., the runtime) per trajectory. This definition is well suited to an experimental procedure where each trajectory is run for a predetermined, fixed time T_{sim} , so T_{all} gives the overall time such an experiment would take. This metric is conservative in the sense that more sophisticated experimental heuristics for predicting when to stop the trajectories earlier than T_{sim} could lead to faster sampling (bringing T_{all} closer to T_{samp}). In addition to T_{all} , we also study the time T_{any} to sample any one of the ground or first-excited configurations, which is similarly defined as the number of trajectories taken to sample any one of the ground or first-excited configurations, multiplied by T_{sim} .

Figure 8 shows how the sampling performance of the

MFB-CIM scales with problem size N . For any given N , we consider 50 problem instances from the SK1 problem class. A representative instance of this problem class can be found in Fig. 8(a), which shows that the nondiagonal elements of the problem matrix $J_{ij} \in \pm 1$. As shown in Fig. 8(b), this problem class has a large total number of degenerate ground and first-excited configurations N_{conf} , which is beneficial for evaluating sampling performance. In this paper, the degenerate ground and first-excited configurations of these problem instances have been identified using the parallel-tempering algorithm [48]. Although parallel tempering is a heuristic algorithm and does not guarantee we identified all ground and first-excited configurations, it has been shown to reliably find the ground energy for the problem instances we consider [36], and, in principle, it is capable of exhaustively finding all the configurations, as the algorithm inherently produces fair samples provided it is run for a sufficiently long time (for the stochastic process to equilibrate). Figure 8(c) shows the distribution of the sampling time T_{all} for various problem sizes. These simulations are per-

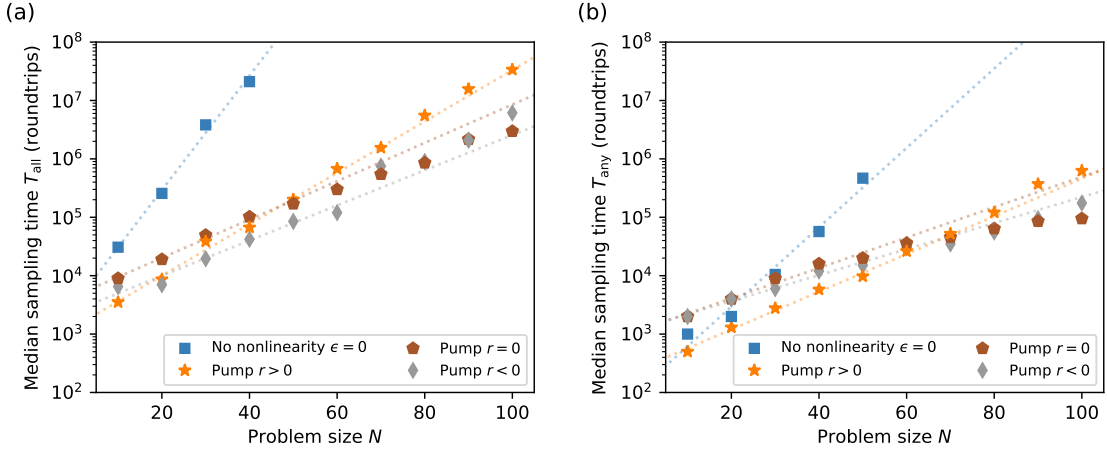


FIG. 9. Sampling performance with various alternative CIM-sampling schemes, as described in Sec. III C. (a) Median time to sample all the ground and first-excited configurations as a function of problem size N . (b) Median time to sample any one of the ground or first-excited configurations as a function of problem size N . For each N , 50 SK1 problem instances are used considered to compute the medians, as done in Fig. 8. In both panels, the dotted lines represent least-squares fits of the median sampling times with respect to problem size. The parameters for each model are specified in Table I.

formed with negative pump parameter ($r < 0$) since this regime was found to perform robustly for the $N = 16$ instance studied in Sec. III C. Additional details about the various model parameters used are shown in Table I. As stated in the table, the key parameters of r and α are stochastically varied from trajectory to trajectory in order to account for problem-dependent variations in the dynamical threshold of the MFB-CIM, which can lead to some problem instances being stuck (an issue easy to detect and correct experimentally). Analogously to Fig. 8(c), Fig. 8(d) shows the distribution of the sampling time T_{any} , and we note the exponential scaling is consistent with prior results for the time required in Ising optimization [7]. On the other hand, studying the scaling of T_{any} *against* that of T_{all} provides insight into the overhead needed to sample many configurations: the difference in the base of the exponent (1.05 vs 1.08) suggests that while the overhead scales exponentially, the penalty (with a base ≈ 1.03) is not especially high. Finally, in Fig. 8(e), we show the scaling of the normalized sampling time $\tilde{T}_{\text{all}}$ (specifically, normalized by the median T_{all} at each N) with respect to N_{conf} . We see that there is correlation between the two quantities and that, for a fixed problem size, the sampling time scales approximately quadratically with respect to the number of configurations.

To put these results into experimental context, the sampling time in roundtrips can also be converted into wall-clock time by multiplying by the roundtrip time of the cavity, which, for a time-multiplexed MFB-CIM, is $\sim N/f_{\text{rep}}$. Thus, for a problem size of $N = 100$ and assuming a source laser with a repetition rate of 10 GHz, Fig. 8(c) indicates that all configurations can be sampled within a median wall-clock time of 60 ms. As a subject for future work, it would be interesting to perform more

	N_{decay}	α	r	η_{esc}
Positive pump	4	$40 + 10z$	$-0.8 + 0.08z$	0.2
No pump	4	$30 + 5z$	0	0.5
Negative pump	1	$4 + 0.6z$	$0.8 + 0.05z$	0.5
No nonlinearity	2	$10 + 2z$	–	0.5

TABLE I. Model parameters used in Sec. IV for simulating the MFB-CIM across different operational modes (see Sec. III C) in studying the large- N scaling of sampling performance. For all simulations, $T_{\text{sim}} = 50N_{\text{decay}}$, and $n_{\text{sat}} = 200$ for all simulations except the case of $\epsilon = 0$, where it is undefined. Here, $z \sim \mathcal{N}(0, 1)$ denotes a standard-normal random variable that perturbs the associated parameter from one trajectory to the next according to the given formula.

thorough benchmark studies to see how these wall-clock times compare to those attained by contemporary algorithms running on conventional digital hardware.

Lastly, we also examine how some alternative modes of operation in the MFB-CIM perform relative to each other. Figure 9 shows, for the models considered in Sec. III C [49], how the medians of the time T_{all} to sample all configurations or T_{any} to sample any configuration scales with problem size N . The parameters used for each model are listed in Table I; due to the large parameter space, the parameters have been heuristically chosen by optimizing over a small set (of size ~ 10), consisting of variations around the optimal parameters found in Sec. III C for $N = 16$. The results show that despite its decent sampling performance in the $N = 16$ case, the coherent-state model scales very poorly, indicating that a linear measurement-feedback protocol in the absence of nonlinearity cannot adequately explain the sampling performance of the MFB-CIM; these results are consistent, for example, with the findings of Ref. [46]. We also ob-

serve that setting the pump parameter to be negative or even zero results in performance that scales similarly to, or arguably even better than, the case of positive pump parameter, both for sampling all configurations as well as any. Considering the experimental advantages of no longer requiring a pump for the system, we expect the $r = 0$ results to be an interesting regime to explore in MFB-CIM sampling experiments.

V. CONCLUSIONS

In this paper, we have formulated a numerically tractable, discrete-time model of the MFB-CIM valid down to the Gaussian-state regime in which quantum noise plays an important role in the system dynamics. Despite being based on the Gaussian-state formalism, however, the model nevertheless captures nonlinear dynamics in the mean and variance of the Gaussian state under experimentally relevant conditions by employing a second-order moment expansion to describe the propagation of the state through the intracavity nonlinearity. The resulting dynamical model is highly general, simultaneously overcoming several restrictions in previously established numerical models for MFB-CIMs: Continuous-time models based on quantum input-output theory only apply to high-finesse CIMs; linear models based on pure Gaussian operations (e.g., squeezing) only apply to CIMs operating below threshold or without optical nonlinearities; and mean-field nonlinear models only apply to high-photon-number CIMs in the classical regime where quantum noise is neglected.

The generality of our model has allowed us to examine the MFB-CIM in the context of a new computational application beyond conventional combinatorial optimization: the dynamical sampling of low-energy Ising spin configurations, driven by quantum noise. We have shown that the sampling behavior first observed in continuous-time Gaussian models of the MFB-CIM [34] persists into the low-finesse regime, carrying the important advantage of increased efficiency by bypassing the diffusive dynamics inherent to the continuous-time limit. We have provided natural parametrizations of our model of relevance to experimental settings, and we have operationally explored sampling performance across a range of these parameters, including pump rate, feedback gain, cavity finesse, and outcoupling efficiency. Using this model, we have explored different operational modes of the MFB-CIM, including negative or zero pump rates, which result in comparable or even enhanced performance, and the absence of optical nonlinearity or quantum noise, both of which result in significant degradation of sampling performance. Due to the compatibility of our model with both existing (low-finesse, high-photon-number) as well as future (quantum-noise-dominated) experimental MFB-CIMs, we expect our numerical results to have immediate implications for the path towards demonstrating efficient Ising sampling on the CIM platform.

In addition to our numerical findings in the context of Ising sampling, this paper also complements and expands upon a longstanding goal of identifying quantum mechanisms and principles of operation in the CIM [29]. The ability to properly treat quantum noise in the Gaussian-state regime using a discrete-time formalism generalizes and validates previous investigations into CIM physics via continuous-time positive-P, truncated-Husimi, and truncated-Wigner SDEs [35, 50], and it also helps clarify the limitations of mean-field models [44, 45] commonly used to study the role of nonlinear dynamics in the large- N limit. Back in the small- N limit, these Gaussian-regime results can act as conceptual semiclassical scaffolding on which to build better understanding of complicated and often unintuitive deep-quantum dynamics. While our focus has been on the measurement-feedback CIM in this paper for the sake of simplicity and experimental relevance, it is straightforward to generalize our approach to describe coherently-coupled CIM networks [6, 9] or potentially even other optical machines like laser networks implementing XY-spin Hamiltonians [51, 52]. In cases where nonlocal entanglement is generated, the cost of representing an entangled Gaussian state only scales as $\mathcal{O}(N^2)$, so our modeling approach can enable intermediate- N numerical studies into the potential role of entanglement in these platforms.

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Appendix A: Continuous-time Gaussian quantum models and the high-finesse limit

As outlined in Sec. IID, the discrete-time model can be reduced to continuous-time models for CIMs derived using conventional quantum optics theory. We first give one example of such a continuous-time quantum model, which produces the Gaussian-state EOMs (24) from the main text. We then analyze each of the discrete map operations described in Sec. IIC, including the nonlinear crystal propagation, to show how (24) can arise in the high-finesse limit of the formalism; in the process we derive the explicit relationships (25) that characterize the scaling of all parameters in our discrete-time model required for the limit to hold. Finally, we present an alternative perspective on this limit in the language of quantum input-output theory, which may also be useful for some readers.

1. Continuous-time Gaussian quantum model

The standard approach to modeling the MFB-CIM is based on input-output theory [33, 53], which describes open quantum systems coupled weakly to a set of external reservoirs. In this formalism, the dynamics are specified by a system Hamiltonian capturing the unitary evolution and a set of Lindblad operators, which describe the interactions of the system with the reservoirs.

For the MFB-CIM, the system of N DOPOs is represented as in the discrete-time case by optical modes with annihilation operators $\hat{a}_i$. The system is coupled to three reservoirs. The first describes unmeasured linear loss and is represented by Lindblad operators $\hat{L}_{\text{loss},i} := \sqrt{2\gamma}\hat{a}_i$, where γ is the field decay rate due to loss. The second describes outcoupling and is represented by Lindblad operators $\hat{L}_{\text{out},i} := \sqrt{2\kappa}\hat{a}_i$, where κ is the field outcoupling rate. Finally, gain saturation is modeled as a two-photon loss corresponding to back-conversion of signal into pump and is represented by Lindblad operators $\hat{L}_{\text{tpl},i} := \sqrt{g}\hat{a}_i^2$, where g is the two-photon loss rate.

The Hamiltonian consists of two coherent effects. The

first is generated by the external pumping of the nonlinear crystal, which gives a contribution of the form $(ip/2)\hat{a}_i^{\dagger 2} + \text{H.c.}$, where p is the field pump rate. The second is generated by external feedback injection, which is a function of the homodyne measurement record obtained from monitoring the output channels $\hat{L}_{\text{out},i}$; we denote this measurement record by

$$m_i(t) := \langle \hat{L}_{\text{out},i} + \hat{L}_{\text{out},i}^\dagger \rangle + \xi_i(t), \quad (\text{A1})$$

where $\xi_i(t)$ is a real-valued standard white noise process with δ -function correlations $\langle \xi_i(t)\xi_j(t') \rangle = \delta_{ij}\delta(t-t')$. Taken together, the system Hamiltonian is given by

$$\hat{H}(t) := \frac{i}{2} \sum_{i=1}^N \left(p\hat{a}_i^{\dagger 2} + \lambda \frac{f_i(t)}{\sqrt{2\kappa}} \hat{a}_i^\dagger \right) + \text{H.c.}, \quad (\text{A2})$$

where $f_i(t) := \sum_j J_{ij}m_j(t)$ is the feedback signal.

Because the measurement records $m_i(t)$ constitute continuous weak measurements of the system state, the dynamics of the system are stochastic and conditional on $m_i(t)$. In standard input-output theory, such dynamics are generated by a stochastic master equation (SME) [54]

$$\begin{aligned} \frac{d\hat{\rho}}{dt} = & -i[\hat{H}(t), \hat{\rho}] + \sum_{i=1}^N \xi_i(t) \mathcal{H}[\hat{L}_{\text{out},i}] \hat{\rho} \\ & + \sum_{i=1}^N \left(\mathcal{D}[\hat{L}_{\text{out},i}] + \mathcal{D}[\hat{L}_{\text{loss},i}] + \mathcal{D}[\hat{L}_{\text{tpl},i}] \right) \hat{\rho}, \end{aligned} \quad (\text{A3})$$

for superoperators $\mathcal{D}[\hat{A}]\hat{\rho} := \hat{A}\hat{\rho}\hat{A}^\dagger - \frac{1}{2}\{\hat{A}^\dagger\hat{A}, \hat{\rho}\}$ and $\mathcal{H}[\hat{A}]\hat{\rho} := \{\hat{A}, \hat{\rho}\} - \langle \hat{A} + \hat{A}^\dagger \rangle \hat{\rho}$.

From the SME, the conditional evolution of any desired observable can be obtained. To establish a correspondence with the discrete-time model, we are particularly interested in the mean and variance of the in-phase quadrature $\hat{q}_i$. In general, the expectation value of an observable $\hat{X}$ has the equation of motion

$$\begin{aligned} \frac{d\langle \hat{X} \rangle}{dt} = & -i \langle [\hat{X}, \hat{H}(t)] \rangle + \sum_{i=1}^N \xi_i(t) \langle \mathcal{H}[\hat{L}_{\text{out},i}^\dagger] \hat{X} \rangle \\ & + \sum_{i=1}^N \left\langle \left(\mathcal{D}[\hat{L}_{\text{out},i}^\dagger] + \mathcal{D}[\hat{L}_{\text{loss},i}^\dagger] + \mathcal{D}[\hat{L}_{\text{tpl},i}^\dagger] \right) \hat{X} \right\rangle. \end{aligned} \quad (\text{A4})$$

We consider $\hat{X}$ to be $\hat{q}_i$ and $\delta\hat{q}_i^2$ to obtain the dynamics of the mean and variance, respectively. As in the discrete-time model, in order to arrive at a closed set of differential equations for the evolution, we assume that the state $\hat{\rho}$ is a Gaussian state at all times. As in the discrete-time model, this Gaussian-state approximation holds when the single-photon nonlinearity is small relative to the linear loss/measurement rates, i.e., $g \ll \kappa + \gamma$. As shown in Appendix B, expectation values of quadrature operators can be evaluated under the Gaussian-state assumption.

Using the procedure outlined there, we arrive at

$$\begin{aligned} \frac{d\langle \hat{q}_i \rangle}{dt} = & (p - \kappa - \gamma) \langle \hat{q}_i \rangle - \frac{g}{2} \langle \hat{q}_i \rangle^3 + \lambda \sum_{j=1}^N J_{ij} \langle \hat{q}_j \rangle \\ & + 2\sqrt{\kappa} \left(\langle \delta\hat{q}_i^2 \rangle - \frac{1}{2} \right) \xi_i + \frac{\lambda}{2\sqrt{\kappa}} \sum_{j=1}^N J_{ij} \xi_j \end{aligned} \quad (\text{A5a})$$

$$\begin{aligned} -\frac{g}{2} \langle \hat{q}_i \rangle \left(3 \langle \delta\hat{q}_i^2 \rangle + \langle \delta\hat{p}_i^2 \rangle - 2 \right), \\ \frac{d\langle \delta\hat{q}_i^2 \rangle}{dt} = & 2(p - \kappa - \gamma) \langle \delta\hat{q}_i^2 \rangle - 4\kappa \left(\langle \delta\hat{q}_i^2 \rangle - \frac{1}{2} \right)^2 \\ & + \kappa + \gamma - 3g \langle \hat{q}_i \rangle^2 \langle \delta\hat{q}_i^2 \rangle + g \langle \hat{q}_i \rangle^2 \\ & - g \left[3 \langle \delta\hat{q}_i^2 \rangle \left(\langle \delta\hat{q}_i^2 \rangle + \frac{1}{3} \langle \delta\hat{p}_i^2 \rangle - 1 \right) - \langle \delta\hat{p}_i^2 \rangle + 1 \right]. \end{aligned} \quad (\text{A5b})$$

For the Gaussian-state approximation to be valid, we require $g \ll \kappa + \gamma$. Thus, terms that scale as g should only be kept if the factor accompanying the g has the capacity to be large. This is for instance satisfied in the saturation term $-g \langle \hat{q}_i \rangle^3$, where a large displacement $\langle \hat{q}_i \rangle$ can make it comparable to the other terms such as $(p - \kappa - \gamma) \langle \hat{q}_i \rangle$. Accordingly, we see that the final terms of both equations in (A5) can in fact be neglected as having loss and measurement $\kappa + \gamma \gg g$ ensures the amount of squeezing/antisqueezing in the MFB-CIM is modest. Removing those terms, we arrive at the simplified continuous-time Gaussian model (24).

2. High-finesse limit of discrete-time dynamics

We now show that continuous-time dynamics of the form (24) can be obtained from the discrete-time model in the high-finesse limit, where each discrete operation in the MFB-CIM only effects a small change to the state.

As discussed in Sec. IID of the main text, the high-finesse limit can be defined by the limit $\delta \rightarrow 0$, where δ scales the parameters of our discrete-time model according to (22). We now consider each of the operations in Sec. IIC and expand each of them up to first order in δ . As usual, the q - and p -quadratures of the dynamics are decoupled, so we only consider the dynamics of $\hat{q}_i$ below.

First, we consider the linear loss at the facets given by (18). Using (5), this produces the mapping

$$\begin{aligned} \langle \hat{q}_i \rangle & \mapsto \sqrt{1 - r_{\text{loss}}^2} \langle \hat{q}_i \rangle \\ & = \left(1 - \frac{1}{2} r_{\text{loss}}^2 \right) \langle \hat{q}_i \rangle + \mathcal{O}(\delta^2), \end{aligned} \quad (\text{A6a})$$

$$\langle \delta\hat{q}_i^2 \rangle \mapsto (1 - r_{\text{loss}}^2) \langle \delta\hat{q}_i^2 \rangle + \frac{1}{2} r_{\text{loss}}^2. \quad (\text{A6b})$$

Since there are two of these facets in a given round-trip, cascading the discrete map twice gives

$$\langle \hat{q}_i \rangle \mapsto (1 - r_{\text{loss}}^2) \langle \hat{q}_i \rangle + \mathcal{O}(\delta^2), \quad (\text{A7a})$$

$$\langle \delta\hat{q}_i^2 \rangle \mapsto (1 - 2r_{\text{loss}}^2) \langle \delta\hat{q}_i^2 \rangle + r_{\text{loss}}^2 + \mathcal{O}(\delta^2). \quad (\text{A7b})$$

Second, we consider the crystal propagation. Since the map (11) requires integrating the nonlinear EOMs (15)

and (16), we use Picard iteration to solve the EOMs while only keeping terms at $\mathcal{O}(\delta)$; the result is an analytic map for $\langle \hat{q}_i \rangle$ and $\langle \delta \hat{q}_i^2 \rangle$ correct up to $\mathcal{O}(\delta)$. With Picard iteration starting from the initial conditions (17), we find that the crystal propagation in the high-finesse limit produces

$$\langle \hat{q}_i \rangle \mapsto \langle \hat{q}_i \rangle + \frac{1}{\sqrt{2}} \beta \epsilon \tau_{\text{nl}} \langle \hat{q}_i \rangle - \frac{1}{8} (\epsilon \tau_{\text{nl}})^2 \langle \hat{q}_i \rangle^3 \quad (\text{A8a})$$

$$\begin{aligned} & - \frac{1}{8} (\epsilon \tau_{\text{nl}})^2 \langle \hat{q}_i \rangle (3 \langle \delta \hat{q}_i^2 \rangle + \langle \delta \hat{p}_i^2 \rangle - 2) + \mathcal{O}(\delta^2), \\ \langle \delta \hat{q}_i^2 \rangle & \mapsto \langle \delta \hat{q}_i^2 \rangle + \sqrt{2} \beta \epsilon \tau_{\text{nl}} \langle \delta \hat{q}_i^2 \rangle \\ & - \frac{3}{4} (\epsilon \tau_{\text{nl}})^2 \langle \hat{q}_i \rangle^2 \langle \delta \hat{q}_i^2 \rangle + \frac{1}{4} (\epsilon \tau_{\text{nl}})^2 \langle \hat{q}_i \rangle^2 \\ & - \frac{1}{4} (\epsilon \tau_{\text{nl}})^2 \langle \delta \hat{q}_i^2 \rangle (\langle \delta \hat{q}_i^2 \rangle - \langle \delta \hat{p}_i^2 \rangle) + \mathcal{O}(\delta^2). \end{aligned} \quad (\text{A8b})$$

We see the last terms in both of the above equations scale as $(\epsilon \tau_{\text{nl}})^2$ and only occur with low powers of the mean $\langle \hat{q}_i \rangle$. We can therefore neglect them following the same argument used above for eliminating the last terms of (A5): Since the Gaussian-state approximation requires $(\epsilon \tau_{\text{nl}})^2 \ll r_{\text{loss}}^2 + r_{\text{out}}^2$, the outcoupling and loss keep the variances close to unity, thus ensuring that these terms remain much smaller than terms at the same order in $\langle \hat{q}_i \rangle$ but associated with r_{loss}^2 and r_{out}^2 .

Third, we consider the measurement process. This consists first of an outcoupling step, which changes the signal state according to

$$\langle \hat{q}_i \rangle \mapsto (1 - \frac{1}{2} r_{\text{out}}^2) \langle \hat{q}_i \rangle + \mathcal{O}(\delta^2), \quad (\text{A9a})$$

$$\langle \delta \hat{q}_i^2 \rangle \mapsto (1 - r_{\text{out}}^2) \langle \delta \hat{q}_i^2 \rangle + \frac{1}{2} r_{\text{out}}^2, \quad (\text{A9b})$$

and also produces a weak correlation between $\hat{q}_i$ and an external mode (labeled here by a subscript h), with mean, variance, and covariance,

$$\langle \hat{q}_h \rangle = r_{\text{out}} \langle \hat{q}_i \rangle, \quad (\text{A10a})$$

$$\langle \delta \hat{q}_h^2 \rangle = \frac{1}{2} + r_{\text{out}}^2 (\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2}), \quad (\text{A10b})$$

$$\langle \delta \hat{q}_i \delta \hat{q}_h \rangle = r_{\text{out}} \sqrt{1 - r_{\text{out}}^2} (\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2}). \quad (\text{A10c})$$

After this, the outcoupled field is measured by homodyne, which by (7) produces a measurement result

$$\begin{aligned} w_i &= \mathcal{N}(\langle \hat{q}_h \rangle, \langle \delta \hat{q}_h^2 \rangle) \\ &= r_{\text{out}} \langle \hat{q}_i \rangle + \frac{1}{\sqrt{2}} [1 + r_{\text{out}}^2 (\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2})] z_i + \mathcal{O}(\delta^2), \end{aligned} \quad (\text{A11})$$

where $z_i \sim \mathcal{N}(0, 1)$ is a standard normal random variable. At the same time, backaction on the internal state by (8) produces the map

$$\begin{aligned} \langle \hat{q}_i \rangle &\mapsto \langle \hat{q}_i \rangle + \left(\frac{w_i - \langle \hat{q}_h \rangle}{\langle \delta \hat{q}_h^2 \rangle} \right) \langle \delta \hat{q}_i \delta \hat{q}_h \rangle \\ &= \langle \hat{q}_i \rangle + \sqrt{2} r_{\text{out}} (\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2}) z_i + \mathcal{O}(\delta^{3/2}), \end{aligned} \quad (\text{A12a})$$

$$\begin{aligned} \langle \delta \hat{q}_i^2 \rangle &\mapsto \langle \delta \hat{q}_i^2 \rangle - \frac{\langle \delta \hat{q}_i \delta \hat{q}_h \rangle^2}{\langle \delta \hat{q}_h^2 \rangle} \\ &= \langle \delta \hat{q}_i^2 \rangle - 2 r_{\text{out}}^2 (\langle \delta \hat{q}_i^2 \rangle - \frac{1}{2})^2 + \mathcal{O}(\delta^2). \end{aligned} \quad (\text{A12b})$$

Finally, we consider injection feedback via (21). Given the measurement results (A11), the displacement we apply is given by $v_i = J_0 \sum_{j=1}^N J_{ij} w_j$, which produces

$$\begin{aligned} \langle \hat{q}_i \rangle &\mapsto \langle \hat{q}_i \rangle + J_0 \sum_{j=1}^N J_{ij} w_j \\ &= \langle \hat{q}_i \rangle + J_0 \sum_{j=1}^N J_{ij} \left(r_{\text{out}} \langle \hat{q}_i \rangle + \frac{1}{\sqrt{2}} z_i \right) + \mathcal{O}(\delta^{3/2}), \end{aligned} \quad (\text{A13})$$

and the variance $\langle \delta \hat{q}_i^2 \rangle$ is unchanged by the feedback.

We can now finally put together all the maps within a single roundtrip by summing up the contributions of (A7), (A8), (A9), (A12), and (A13) up to first order in δ . The resulting updated state described by $\langle \hat{q}_i \rangle'$ and $\langle \delta \hat{q}_i^2 \rangle'$ can then be substituted into the definitions (23). Then by imposing conditions related to the Gaussian-state approximation (due to $(\epsilon \tau_{\text{nl}})^2 \ll r_{\text{loss}}^2 + r_{\text{out}}^2$) as was also done in the derivation of (24), we finally arrive at EOMs identical to (24), *provided* we utilize the relationships explicitly given in (25).

3. Quantum input-output approach

Finally, as an alternative to the above approach where the correspondence between continuous and discrete time is made via manipulation of the (c-number) means and variances, it is also possible to arrive at the same conclusions using a quantum input-output analysis of the crystal Hamiltonian (10) as well. For example, on one roundtrip, the crystal implements a unitary operation

$$\hat{U}_{\text{nl}} := e^{-i \hat{H}_{\text{nl}} \tau_{\text{nl}}} = \exp \left(\frac{\epsilon \tau_{\text{nl}}}{2} \sum_i \hat{b}_i \hat{a}_i^{\dagger 2} - \text{H.c.} \right), \quad (\text{A14})$$

where $\hat{H}_{\text{nl}} := \sum_i \hat{H}_{\text{nl}}^{(i)}$. We can decompose the pump operator as the sum of a coherent-excitation part and a quantum noise part via $\hat{b}_i = \beta_i / \sqrt{2} + \delta \hat{b}_i$, allowing us to treat the parametric amplification and the nonlinear parametric quantum fluctuations separately. With this substitution,

$$\hat{U}_{\text{nl}} = \exp \left(\frac{\beta \epsilon \tau_{\text{nl}}}{2 \sqrt{2}} \sum_i \hat{a}_i^{\dagger 2} + \frac{\epsilon \tau_{\text{nl}}}{2} \sum_i \delta \hat{b}_i \hat{a}_i^{\dagger 2} - \text{H.c.} \right). \quad (\text{A15})$$

In the high-finesse limit where $\beta^2 \sim (\epsilon \tau_{\text{nl}})^2 \sim \Delta t \sim \delta$, this unitary evolution can be made compatible with a discrete-time picture of the dynamics if we Trotterize [43] the above unitary over one roundtrip time by writing

$$\hat{U}_{\text{nl}} = \exp \left(-i \hat{H}_{\text{sqz}} \Delta t \right) \exp \left(-i \hat{H}_{\text{tpl}} \Delta t \right) + \mathcal{O}(\delta^{3/2}), \quad (\text{A16a})$$

where the first exponential effects a rotation $\mathcal{O}(\delta)$ and is generated by a squeezing Hamiltonian

$$\hat{H}_{\text{sqz}} := \frac{i}{2} \underbrace{\frac{\beta \epsilon \tau_{\text{nl}}}{\sqrt{2} \Delta t}}_p \sum_i \hat{a}_i^{\dagger 2} + \text{H.c.}, \quad (\text{A16b})$$

while the second exponential effects a rotation $\mathcal{O}(\delta^{1/2})$ and is generated by an interaction Hamiltonian that we can write as

$$\hat{H}_{\text{tpl}} := i \underbrace{\frac{\epsilon \tau_{\text{nl}}}{2\sqrt{\Delta t}}}_{\sqrt{g}} \sum_i \hat{b}_i^{(\text{in},t)} \hat{a}_i^{\dagger 2} + \text{H.c.}, \quad (\text{A16c})$$

where in the limit $\Delta t \sim \delta \rightarrow 0$, the quantum white-noise operators $\hat{b}_i^{(\text{in},t)} := \delta \hat{b}_i / \sqrt{\Delta t}$ have Dirac-delta commutation relations, i.e., $[\hat{b}_i^{(\text{in},t)}, \hat{b}_{i'}^{(\text{in},t')\dagger}] = \delta_{i,i'} \delta(t-t')$.

In a coarse-grained continuous-time theory over many roundtrips, (A16b) is precisely the gain/squeezing part of the continuous-time system Hamiltonian (23), while (A16c) is an input-output system-reservoir interaction Hamiltonian that formally defines the continuous-time Lindblad operator $\hat{L}_{\text{tpl},i}$ in the continuous-time model. This process of Trotterizing discrete-time operations can also be applied to all the linear operations (loss, outcoupling, measurement, and feedback) as well.

Appendix B: Evaluating expectation values of quadrature operators

Here we outline how to evaluate expectation values of an operator of the form $\hat{q}^r \hat{p}^m$ on a single-mode Gaussian state. We include this section largely for pedagogical purposes as we found these results are typically presented in more general, and hence less accessible, terms than necessary for the specific scenario we consider [55].

First we note the Weyl-ordered (i.e., symmetrically ordered) expression for the operator is given by

$$\hat{q}^r \hat{p}^m = \sum_{j=0}^{\min(r,m)} \left(\frac{i}{2}\right)^2 \binom{r}{j} \binom{m}{j} j! (\hat{q}^{r-j} \hat{p}^{m-j})_{\text{W}}, \quad (\text{B1})$$

where $(\cdot)_{\text{W}}$ denotes operators that are Weyl-ordered [55]. With the operators in Weyl form, the expectation value can be evaluated by a phase-space integral

$$\langle (\hat{q}^{r-j} \hat{p}^{m-j})_{\text{W}} \rangle = \int_{\mathbb{R}^2} q^{r-j} p^{m-j} W(q, p) dq dp, \quad (\text{B2})$$

where $W(q, p)$ is the Wigner function for the quantum state. For a Gaussian state, the Wigner function is given by a multivariate normal distribution:

$$W(z) = \frac{1}{2\pi \det \Sigma} \exp \left(-\frac{1}{2} (z - \mu)^T \Sigma^{-1} (z - \mu) \right), \quad (\text{B3})$$

where $z = (q, p)^T$ and μ and Σ are the mean vector and covariance matrix of the state, as defined in (1).

Appendix C: Crystal propagation EOMs

Here we present the full equations of motion for the evolution of the Gaussian moments for the joint signal-pump state as it propagates through the crystal according to (10), using the procedure described in Sec. II B.

The full mean-field equations of motion are

$$\begin{aligned} \frac{d \langle \hat{x}_i \rangle}{d(\epsilon \tau)} &= \langle \hat{x}_b \rangle \langle \hat{x}_i \rangle + \langle \hat{y}_b \rangle \langle \hat{y}_i \rangle + \langle \delta \hat{x}_b \delta \hat{x}_i + \delta \hat{y}_b \delta \hat{y}_i \rangle, \\ \frac{d \langle \hat{x}_b \rangle}{d(\epsilon \tau)} &= -\frac{1}{2} (\langle \hat{x}_i \rangle^2 - \langle \hat{y}_i \rangle^2) - \frac{1}{2} \langle \delta \hat{x}_i^2 - \delta \hat{y}_i^2 \rangle, \\ \frac{d \langle \hat{y}_i \rangle}{d(\epsilon \tau)} &= \langle \hat{y}_b \rangle \langle \hat{x}_i \rangle - \langle \hat{x}_b \rangle \langle \hat{y}_i \rangle + \langle \delta \hat{y}_b \delta \hat{x}_i - \delta \hat{x}_b \delta \hat{y}_i \rangle, \\ \frac{d \langle \hat{y}_b \rangle}{d(\epsilon \tau)} &= -\langle \hat{x}_i \rangle \langle \hat{y}_i \rangle - \frac{1}{2} \langle \delta \hat{x}_i \delta \hat{y}_i + \delta \hat{y}_i \delta \hat{x}_i \rangle, \end{aligned}$$

while for the covariances, we have

$$\begin{aligned} \frac{d \langle \delta \hat{x}_i^2 \rangle}{d(\epsilon \tau)} &= +2 \langle \hat{x}_b \rangle \langle \delta \hat{x}_i^2 \rangle + 2 \langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle + 2 \langle \hat{y}_i \rangle \langle \delta \hat{y}_b \delta \hat{x}_i \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{x}_i \delta \hat{y}_i + \delta \hat{y}_i \delta \hat{x}_i \rangle, \\ \frac{d \langle \delta \hat{y}_i^2 \rangle}{d(\epsilon \tau)} &= -2 \langle \hat{x}_b \rangle \langle \delta \hat{y}_i^2 \rangle + 2 \langle \hat{x}_i \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle - 2 \langle \hat{y}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_i \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{x}_i \delta \hat{y}_i + \delta \hat{y}_i \delta \hat{x}_i \rangle, \\ \frac{d \langle \delta \hat{x}_b^2 \rangle}{d(\epsilon \tau)} &= -2 \langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle - 2 \langle \hat{y}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_i \rangle, \\ \frac{d \langle \delta \hat{y}_b^2 \rangle}{d(\epsilon \tau)} &= -2 \langle \hat{x}_i \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle - 2 \langle \hat{y}_i \rangle \langle \delta \hat{y}_b \delta \hat{x}_i \rangle, \\ \frac{d \langle \delta \hat{x}_b \delta \hat{x}_i \rangle}{d(\epsilon \tau)} &= +\langle \hat{x}_b \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle + \langle \hat{x}_i \rangle \langle \delta \hat{x}_b^2 - \delta \hat{x}_i^2 \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{x}_b \delta \hat{y}_i \rangle + \langle \hat{y}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_i + \delta \hat{y}_i \delta \hat{x}_i \rangle, \\ \frac{d \langle \delta \hat{y}_b \delta \hat{y}_i \rangle}{d(\epsilon \tau)} &= -\langle \hat{x}_b \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle + \langle \hat{x}_i \rangle \langle \delta \hat{y}_b^2 - \delta \hat{y}_i^2 \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{y}_b \delta \hat{x}_i \rangle - \langle \hat{y}_i \rangle \langle \delta \hat{x}_i \delta \hat{y}_i + \delta \hat{y}_i \delta \hat{x}_i \rangle, \\ \frac{d \langle \delta \hat{x}_i \delta \hat{y}_i \rangle}{d(\epsilon \tau)} &= +\langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_i + \delta \hat{y}_b \delta \hat{x}_i \rangle - \langle \hat{y}_i \rangle \langle \delta \hat{x}_b \delta \hat{x}_i - \delta \hat{y}_b \delta \hat{y}_i \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{x}_i^2 + \delta \hat{y}_i^2 \rangle, \end{aligned}$$

$$\begin{aligned}
\frac{d \langle \delta \hat{x}_b \delta \hat{y}_b \rangle}{d(\epsilon \tau)} &= -\langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_i + \delta \hat{y}_b \delta \hat{x}_i \rangle - \langle \hat{y}_i \rangle \langle \delta \hat{x}_b \delta \hat{x}_i - \delta \hat{y}_b \delta \hat{y}_i \rangle, \\
\frac{d \langle \delta \hat{x}_b \delta \hat{y}_i \rangle}{d(\epsilon \tau)} &= -\langle \hat{x}_b \rangle \langle \delta \hat{x}_b \delta \hat{y}_i \rangle + \langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_b - \delta \hat{x}_i \delta \hat{y}_i \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{x}_b \delta \hat{x}_i \rangle - \langle \hat{y}_i \rangle \langle \delta \hat{x}_b^2 - \delta \hat{y}_i^2 \rangle, \\
\frac{d \langle \delta \hat{y}_b \delta \hat{x}_i \rangle}{d(\epsilon \tau)} &= +\langle \hat{x}_b \rangle \langle \delta \hat{y}_b \delta \hat{x}_i \rangle + \langle \hat{x}_i \rangle \langle \delta \hat{x}_b \delta \hat{y}_b - \delta \hat{x}_i \delta \hat{y}_i \rangle + \langle \hat{y}_b \rangle \langle \delta \hat{y}_b \delta \hat{y}_i \rangle + \langle \hat{y}_i \rangle \langle \delta \hat{y}_b^2 - \delta \hat{x}_i^2 \rangle.
\end{aligned}$$
